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ONIUM SALT, CHEMICALLY AMPLIFIED RESIST COMPOSITION, AND PATTERNING PROCESS

Abstract

An onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure is provided. A chemically amplified resist composition comprising the onium salt has satisfactory solvent solubility, high sensitivity, and high contrast, and forms a resist film with improved lithography properties such as EL and LWR.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No. 2024-031892 filed in Japan on Mar. 4, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] This invention relates to an onium salt, a chemically amplified resist composition, and a pattern forming process.

BACKGROUND ART

[0003] While a number of recent efforts are being made to achieve a finer pattern rule in the drive for higher integration and operating speeds in LSIs, DUV and EUV lithography processes are thought to hold particular promise as the next generation in microfabrication technology. In particular, photolithography using an ArF excimer laser is requisite to the micropatterning technique capable of achieving a feature size of 0.13 μm or less.

[0004] The ArF lithography started partial use from the fabrication of 130-nm node devices and became the main lithography since 90-nm node devices. Although lithography using F.sub.2 laser (wavelength 157 nm) was initially thought promising as the next lithography for 45-nm node devices, its development was retarded by several problems. A highlight was suddenly placed on the ArF immersion lithography that introduces a liquid having a higher refractive index than air (e.g., water, ethylene glycol, glycerin) between the projection lens and the wafer, allowing the projection lens to be designed to a numerical aperture (NA) of 1.0 or higher and achieving a higher resolution. See Non-Patent Document 1. The ArF immersion lithography is now implemented on the commercial stage. The immersion lithography requires a resist material which is substantially insoluble in water.

[0005] In the photolithography using an ArF excimer laser (wavelength 193 nm), a high sensitivity resist material capable of achieving a high resolution at a small dose of exposure is needed to prevent the degradation of precise and expensive optical system materials. Among several measures for providing the high sensitivity resist material, the most common is to select each component which is highly transparent at the wavelength of 193 nm. For example, polymers of acrylic acid and derivatives thereof, norbornene-maleic anhydride alternating copolymers, polynorbornene, ring-opening metathesis polymerization (ROMP) polymers, and hydrogenated ROMP polymers have been proposed as the base resin. This choice is effective to some extent in that the transparency of a resin alone is increased.

[0006] Recently a highlight is put on the negative tone resist adapted for organic solvent development as well as the positive tone resist adapted for aqueous alkaline development. It would be desirable if a very fine hole pattern, which is not achievable with the positive tone, is resolvable through negative tone exposure. To this end, a positive resist material featuring a high resolution is subjected to organic solvent development to form a negative pattern. An attempt to double a resolution by combining two developments, aqueous alkali development and organic solvent development is under study. As the ArF resist material for negative tone development with organic solvent, positive ArF resist compositions of the prior art design may be used. Such pattern forming processes are described in Patent Documents 1 to 3.

[0007] To meet the current rapid progress of microfabrication technology, development efforts are put on not only the process, but also the resist material. Studies have also been made on photoacid generators (PAGs). Commonly used are sulfonium salts of triphenylsulfonium cations with perfluoroalkanesulfonic acid anions. These salts generate perfluoroalkanesulfonic acids, especially perfluorooctanesulfonic acid (PFOS), which are considered problematic with respect to their non-degradability, biological concentration and toxicity. It is rather restricted to apply these salts to the resist material. Instead, PAGs capable of generating perfluorobutanesulfonic acid are currently used, but are awkward to achieve a high resolution because of substantial diffusion of the generated acid in the resist material. To address the problem, partially fluorinated alkane sulfonic acids and salts thereof are developed. For instance, Patent Document 1 describes the prior art PAGs capable of generating α,α -difluoroalkanesulfonic acid, such as di(4-tert-butylphenyl)iodonium 1,1-difluoro-2-(1-naphthyl)ethanesulfonate and PAGs capable of generating $\alpha,\alpha,\beta,\beta$ -tetrafluoroalkanesulfonic acid. Despite a reduced degree of fluorine substitution, these PAGs still have the following problems. Since they do not have a decomposable substituent group such as ester structure, they are unsatisfactory from the aspect of environmental safety or ease of decomposition. The molecular design to change the size of alkanesulfonic acid is limited. Fluorine-containing starting reactants are expensive.

[0008] As the circuit line width is reduced, the degradation of contrast by acid diffusion becomes more serious for the resist material. The reason is that the pattern feature size is approaching the diffusion length of acid. This invites a lowering of mask fidelity and a degradation of pattern rectangularity because a dimensional shift on wafer (known as mask error factor (MEF)) relative to a dimensional shift on mask is exaggerated. Accordingly, to gain more benefits from a reduction of exposure light wavelength and an increase of lens NA, the resist material is required to increase a dissolution contrast or restrain acid diffusion, as compared with the prior art materials. One approach is to lower the bake temperature for suppressing acid diffusion and hence, improving MEF. A low bake temperature, however, inevitably leads to a low sensitivity.

[0009] Incorporating a bulky substituent or polar group into PAG is effective for suppressing acid diffusion. Patent Document 4 discloses a PAG capable of generating 2-acyloxy-1,1,3,3,3-pentafluoropropane-1-sulfonic acid which is fully soluble and stable in resist solvents and allows for a wide span of molecular design. In particular, a PAG having a bulky substituent incorporated therein or capable of generating 2-(1-adamantyloxy)-1,1,3,3,3-pentafluoropropane-1-sulfonic acid is characterized by slow acid diffusion. Patent Documents 5 to 7 describe PAGs having fused ring lactone, sultone or thiolactone incorporated as the polar group. Although some improvement in performance is observed due to the acid diffusion suppressing effect of the polar group incorporated, they are still insufficient in precise control of acid diffusion. Their lithography performance is unsatisfactory when evaluated totally in terms of MEF, pattern profile and sensitivity.

[0010] Incorporating a polar group into an anion of PAG is effective for suppressing acid diffusion, but disadvantageous from the standpoint of solvent solubility. Attempting to improve solvent solubility, Patent Documents 8 and 9 propose to incorporate an alicyclic group into a cation moiety of a PAG. Specifically, a cyclohexane or adamantane ring is incorporated. While incorporating such an alicyclic group achieves an improvement in solubility, a relatively large number of carbon atoms is necessary to insure a satisfactory solubility. This means that the molecular structure of PAG becomes bulky, causing to degrade lithography properties such as LWR and CDU in forming small-size patterns.

[0011] Patent Document 10 describes a PAG containing an anion having an aromatic fused ring derived from anthracene and adapted to generate a fluoroalkanesulfonic acid. Although the lithography performance is improved to some extent, the alkanesulfonic acid structure lacks robustness and its influence to the environment and human body is concerned as viewed from the current circumstance where organic fluorine compounds within the class of PFAS are regulated. To

meet the demand for further miniaturization, it is crucial to develop a novel PAG. There is the desire to have a PAG capable of fully suppressing acid diffusion, achieving high solvent solubility, and improving lithography properties.

CITATION LIST

[0012] Patent Document 1: JP-A 2008-281974 [0013] Patent Document 2: JP-A 2008-281975 [0014] Patent Document 3: JP 4554665 [0015] Patent Document 4: JP-A 2007-145797 [0016] Patent Document 5: JP 5061484 [0017] Patent Document 6: JP-A 2016-147879 [0018] Patent Document 7: JP-A 2015-063472 [0019] Patent Document 8: JP 5573098 [0020] Patent Document 9: JP 6461919 Patent Document 10: JP 7109178 [0021] Non-Patent Document 1: Journal of Photopolymer Science and Technology, Vol. 17, No. 4, p. 587-601 (2004)

SUMMARY OF THE INVENTION

[0022] While it is recently demanded to form resist patterns at a high resolution, a resist composition using a PAG of conventional onium salt type fails to fully suppress acid diffusion. As a result, lithography properties such as contrast, MEF and LWR are degraded.

[0023] An object of the invention is to provide an onium salt and a chemically amplified resist composition comprising the same, the resist composition having a high solvent solubility, high sensitivity and high contrast, and forming a resist film with improved lithography properties such as contrast, EL and LWR when processed by photolithography using high-energy radiation such as KrF or ArF excimer laser, EB or EUV; and a pattern forming process using the resist composition.

[0024] The inventor has found that an onium salt containing a sulfonic acid anion having a substituted fused ring structure and an aromatic sulfonic acid structure has a high solvent solubility and that a chemically amplified resist composition comprising the onium salt as a photoacid generator exhibits a high sensitivity, high contrast, acid diffusion inhibition, and improved lithography properties such as EL and LWR, and is quite effective for forming small-size patterns.

[0025] In one aspect, the invention provides an onium salt having the formula (1).

##STR00001##

[0026] Herein R.sup.1 to R.sup.12 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom,

[0027] one of R.sup.13 and R.sup.14 is a group having a partial structure having the formula (1a) and the other is hydrogen, halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, [0028] at least two of R.sup.1 to R.sup.14 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and [0029] Z.sup.+ is an onium cation,

##STR00002## [0030] wherein m₁ is 0 or 1, m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1, m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1, m₂+m₃ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1, [0031] R.sup.F is fluorine, a C.sub.1-C.sub.6 fluorinated alkyl group, C.sub.1-C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group, [0032] R.sup.15 is a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, a plurality of R.sup.15 may bond together to form a ring with the carbon atoms to which they are attached when m₃ is 2 or more, [0033] L.sup.A and L.sup.B are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and [0034] X.sup.L is a single bond or a C.sub.1-C.sub.40 hydrocarbylene group which may contain a heteroatom.

[0035] Preferably, the onium salt has the formula (1A):

##STR00003## [0036] wherein R.sup.1 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above.

[0037] More preferably, the onium salt has the formula (1B):

##STR00004## [0038] wherein R.sup.5, R.sup.10 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m₁ to m₃, and Z.sup.+ are as defined above, [0039] m₄ and m₅ are each independently an integer of 0 to 4, [0040] R.sup.16 and R.sup.17 are each independently hydrogen, halogen, or a

C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom; when m4 is 2 or more, a plurality of R.sup.16 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m5 is 2 or more, a plurality of R.sup.17 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

[0041] Even more preferably, the onium salt has the formula (1C):

##STR00005## [0042] wherein R.sup.5, R.sup.10 to R.sup.13, R.sup.16, R.sup.17, L.sup.A, X.sup.L, m4, m5, and Z.sup.+ are as defined above.

[0043] In one preferred embodiment, Z.sup.+ is a sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2):

##STR00006## [0044] wherein R.sup.ct1 to R.sup.ct5 are each independently halogen or a C.sub.1-C.sub.30 hydrocarbyl group which may contain a heteroatom, R.sup.ct1 and R.sup.ct2 may bond together to form a ring with the sulfur atom to which they are attached.

[0045] In another aspect, the invention provides a photoacid generator in the form of the onium salt defined herein.

[0046] In a further aspect, the invention provides a chemically amplified resist composition comprising the photoacid generator defined herein.

[0047] In one preferred embodiment, the resist composition comprises a base polymer comprising repeat units having the formula (a1).

##STR00007##

[0048] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0049] X.sup.1 is a single bond, phenylene, naphthylene or *—C(=O)—O—X.sup.11—, the phenylene or naphthylene group may be substituted with an optionally fluorinated C.sub.1-C.sub.10 alkoxy moiety or halogen, X.sup.11 is a C.sub.1-C.sub.10 saturated hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, a phenylene group or a naphthylene group, * designates a point of attachment to the carbon atom in the backbone, and [0050] AL.sup.1 is an acid labile group.

[0051] In one preferred embodiment, the base polymer further comprises repeat units having the formula (a2).

##STR00008##

[0052] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0053] X.sup.2 is a single bond or *—C(=O)—O—, * designates a point of attachment to the carbon atom in the backbone,

[0054] R.sup.21 is halogen, cyano, a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, C.sub.1-C.sub.20 hydrocarbyloxy group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyl group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyloxy group which may contain a heteroatom, or C.sub.2-C.sub.20 hydrocarbyloxycarbonyl group which may contain a heteroatom, [0055] AL.sup.2 is an acid labile group, and [0056] a is an integer of 0 to 4.

[0057] In one preferred embodiment, the base polymer further comprises repeat units having the formula (b1) or (b2).

##STR00009##

[0058] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0059] Y.sup.1 is a single bond or *—C(=O)—O—, * designates a point of attachment to the carbon atom in the backbone,

[0060] R.sup.31 is hydrogen, or a C.sub.1-C.sub.20 group containing at least one structure selected from hydroxy exclusive of phenolic hydroxy, cyano, carbonyl, carboxy, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, and carboxylic anhydride (—C(=O)—O—C(=O)—), [0061] R.sup.32 is halogen, hydroxy, nitro, a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, C.sub.1-C.sub.20 hydrocarbyloxy group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyl group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyloxy group which may contain a heteroatom, or

C.sub.2-C.sub.20 hydrocarbyloxycarbonyl group which may contain a heteroatom, [0062] b is an integer of 1 to 4, c is an integer of 0 to 4, and b+c is from 1 to 5.

[0063] In one preferred embodiment, the base polymer further comprises repeat units of at least one type selected from repeat units having the formula (c1), repeat units having the formula (c2), repeat units having the formula (c3), and repeat units having the formula (c4).

##STR00010##

[0064] Herein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, [0065] Z.sup.1 is a single bond or phenylene group, [0066] Z.sup.2 is *—C(=O)—O—Z.sup.21—, *—C(=O)—NH—Z.sup.21—, or *—O—Z.sup.21—, Z.sup.21 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group or a divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, [0067] Z.sup.3 is each independently a single bond, phenylene, naphthylene, or *—C(=O)—O—Z.sup.31—, Z.sup.31 is a C.sub.1-C.sub.10 aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, or phenylene or naphthylene group, [0068] Z.sup.4 is each independently a single bond, **—Z.sup.41—C(=O)—O—, **—C(=O)—NH—Z.sup.41—, or —O—Z.sup.41—, Z.sup.41 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, [0069] Z.sup.5 is each independently a single bond, *—Z.sup.5—C(=O)—O—, *—C(=O)—NH—Z.sup.5—, or *—O—Z.sup.51, Z.sup.51 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, [0070] Z.sup.6 is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *—C(=O)—O—Z.sup.61—, *—C(=O)—N(H)—Z.sup.61—, or *—O—Z.sup.61, Z.sup.61 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, [0071] * designates a point of attachment to the carbon atom in the backbone, ** designates a point of attachment to Z.sup.3, [0072] R.sup.41 and R.sup.42 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.41 and R.sup.42 may bond together to form a ring with the sulfur atom to which they are attached, [0073] L.sup.1 is a single bond, ether bond, ester bond, carbonyl group, sulfonate ester bond, carbonate bond or carbamate bond, [0074] Rf.sup.1 and Rf.sup.2 are each independently fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, [0075] Rf.sup.3 and Rf.sup.4 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, [0076] Rf.sup.5 and Rf.sup.6 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, excluding that all Rf.sup.5 and Rf.sup.6 are hydrogen at the same time, [0077] M.sup.— is a non-nucleophilic counter ion, [0078] A.sup.+ is an onium cation, and [0079] d is an integer of 0 to 3. [0080] The resist composition may further comprise an organic solvent, a quencher, a photoacid generator other than the photoacid generator defined herein, and/or a surfactant.

[0081] In a still further aspect, the invention provides a pattern forming process comprising the steps of applying the chemically amplified resist composition defined herein onto a substrate to form a resist film thereon, exposing the resist film to high-energy radiation, and developing the exposed resist film in a developer.

[0082] Typically, the high-energy radiation is KrF excimer laser radiation, ArF excimer laser radiation, EB or EUV of wavelength 3 to 15 nm.

Advantageous Effects of Invention

[0083] When the chemically amplified resist composition comprising the onium salt as a photoacid generator is processed by lithography, resist patterns having a high sensitivity, acid diffusion inhibition, and improved properties including MEF and LWR can be formed. The risk of pattern collapse during formation of small-size patterns is minimized.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0084] The only FIGURE, FIG. 1 is a diagram showing the ¹H-NMR spectrum of onium salt PAG-1 synthesized in Example 1-1.

DETAILED DESCRIPTION OF THE INVENTION

[0085] As used herein, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. “Optional” or “optionally” means that the subsequently described event or circumstances may or may not occur, and that description includes instances where the event or circumstance occurs and instances where it does not. The notation (C_n-C_m) means a group containing from n to m carbon atoms per group. In chemical formulae, Me stands for methyl, and Ac for acetyl. The broken line (---) and the asterisk (*) designate a valence bond or point of attachment. As used herein, the term “halogenated” refers to a halogen-substituted or halogen-containing compound or group. The terms “group” and “moiety” are interchangeable.

[0086] The abbreviations and acronyms have the following meaning. [0087] EB: electron beam

[0088] EUV: extreme ultraviolet [0089] Mw: weight average molecular weight [0090] Mn: number average molecular weight [0091] Mw/Mn: molecular weight dispersity [0092] GPC: gel

permeation chromatography [0093] PEB: post-exposure bake [0094] PAG: photoacid generator

[0095] EL: exposure latitude [0096] LWR: line width roughness [0097] MEF: mask error factor

[0098] CDU: critical dimension uniformity [0099] DOF: depth of focus

[0100] It is understood that for some structures represented by chemical formulae, there can exist enantiomers and diastereomers because of the presence of asymmetric carbon atoms. In such a case, a single formula collectively represents all such isomers. The isomers may be used alone or in admixture.

[Onium Salt]

[0101] The invention provides an onium salt having the formula (1).

##STR00011##

[0102] In formula (1), R¹ to R¹² are each independently hydrogen, halogen, or a C₁-C₂₀ hydrocarbyl group which may contain a heteroatom. One of R¹³ and R¹⁴ is a group having a partial structure having the formula (1a) shown later while the other is hydrogen, halogen or a C₁-C₂₀ hydrocarbyl group which may contain a heteroatom.

[0103] Suitable halogen atoms include fluorine, chlorine, bromine, and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C₁-C₂₀ alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-octyl, n-nonyl, n-decyl, undecyl, dodecyl, tridecyl, tetradecyl, pentadecyl, heptadecyl, octadecyl, nonadecyl, and icosyl; C₃-C₂₀ cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, and adamantyl; C₂-C₂₀ alkenyl groups such as vinyl, allyl, propenyl, butenyl, and hexenyl; C₃-C₂₀ cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C₆-C₂₀ aryl groups such as phenyl and naphthyl; C₇-C₂₀ aralkyl groups such as benzyl, 1-phenylethyl, and 2-phenylethyl, and combinations thereof. Inter alia, aryl groups are preferred. In the hydrocarbyl group, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH₂— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, cyano, fluorine, chlorine, bromine, iodine, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0104] At least two of R¹ to R¹⁴ may bond together to form a ring with the carbon to which they are attached, or the carbon to which they are attached and the intervenient carbon. Examples of the ring thus formed include aliphatic rings such as cyclopropane, cyclobutane,

cyclopentane, cyclohexane, norbornane, and adamantane rings and aromatic rings such as benzene, naphthalene and anthracene rings. In the ring, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH₂— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the ring may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Of the rings, aromatic rings are preferred, and benzene rings are more preferred.

[0105] In formula (1), one of R¹³ and R¹⁴ is a group having a partial structure having the following formula (1a). It is preferred from the aspect of synthesis that R¹⁴ be a group having a partial structure having the formula (1a).

##STR00012##

[0106] In formula (1a), m₁ is 0 or 1. The relevant structure is a benzene ring when m₁=0, and a naphthalene ring when m₁=1. The benzene ring corresponding to m₁=0 is preferred from the aspect of solvent solubility. The subscript m₂ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1. Preferably, m₂ is 4 when m₁=0, and m₂ is 4, 5 or 6 when m₁=1. The subscript m₃ is an integer of 0 to 3 when m₁=0 and an integer of 0 to 5 when m₁=1. The sum m₂+m₃ is an integer of 0 to 4 when m₁=0 and an integer of 0 to 6 when m₁=1.

[0107] In formula (1a), R^F is fluorine, a C₁-C₆ fluorinated alkyl group, C₁-C₆ fluorinated alkoxy group, or C₁-C₆ fluorinated alkylthio group. R^F is preferably fluorine, trifluoromethyl, difluoromethyl, trifluoromethoxy, difluoromethoxy, trifluoromethylthio or difluoromethylthio, more preferably fluorine, trifluoromethyl or trifluoromethoxy. When fluorine or a fluorinated substituent group is contained, the acid strength of the generated acid is so enhanced due to the electron attractive effect that deprotection reaction of acid labile groups such as tertiary ester or tertiary ether groups may smoothly take place. When m₂ is 2, 3 or 4, a plurality of R^F may be identical or different.

[0108] In formula (1a), R¹⁵ is a C₁-C₂₀ hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl group R¹, but not limited thereto. When m₃ is 2 or more, a plurality of R¹⁵ may be identical or different. When m₃ is 2 or more, a plurality of R¹⁵ may bond together to form a ring with the carbon atoms to which they are attached. The ring is preferably a 5 to 8-membered ring.

[0109] In formula (1a), L^A and L^B are each independently a single bond, ether bond, ester bond, amide bond, sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond. Inter alia, a single bond, ether bond or ester bond is preferred.

[0110] In formula (1a), X^L is a single bond or a C₁-C₄₀ hydrocarbylene group which may contain a heteroatom. The hydrocarbylene group may be straight, branched or cyclic and examples thereof include alkanediyl groups, cyclic saturated hydrocarbylene groups, and arylene groups. Suitable heteroatoms include oxygen, nitrogen and sulfur.

[0111] Examples of the C₁-C₄₀ hydrocarbylene group which may contain a heteroatom, represented by X^L, are shown below, but not limited thereto. Herein, * designates a point of attachment to L^A or L^B.

##STR00013## ##STR00014##

[0112] Of these, X^L₀ to X^L₂₂ and X^L₄₇ to X^L₅₈ are preferred.

[0113] Of the onium salts having formula (1), those having the formula (1A) are preferred.

##STR00015##

[0114] Herein R¹ to R¹³, R¹, L^A, X^L, R^F, m₁ to m₃, and Z⁺ are as defined above.

[0115] Of the onium salts having formula (1), those having the formula (1B) are more preferred.

##STR00016##

[0116] In formula (1B), R.sup.5, R.sup.10 to R.sup.13, R.sup.15, L.sup.A, X.sup.L, R.sup.F, m1 to m3, and Z.sup.+ are as defined above.

[0117] In formula (1B), m4 and m5 are each independently an integer of 0 to 4. From the aspect of reactant availability, m4 and m5 each are preferably 0, 1 or 2.

[0118] In formula (1B), R.sup.16 and R.sup.17 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. Examples of the halogen and hydrocarbyl group are as exemplified above for the halogen and hydrocarbyl group represented by R.sup.1 to R.sup.14.

[0119] When m4 is 2 or more, a plurality of R.sup.16 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon. When m5 is 2 or more, a plurality of R.sup.17 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon. Examples of the ring are as exemplified above for the ring that is formed by at least two of R.sup.1 to R.sup.14.

[0120] Of the onium salts having formula (1B), those having the formula (1C) are even more preferred.

##STR00017##

[0121] In formula (1C), R.sup.5, R.sup.10 to R.sup.13, R.sup.16, R.sup.17, L.sup.A, X.sup.L, m4, m5, and Z.sup.+ are as defined above.

[0122] Examples of the anion in the onium salt having formula (1) are shown below, but not limited thereto.

##STR00018## ##STR00019## ##STR00020## ##STR00021## ##STR00022## ##STR00023##
##STR00024## ##STR00025## ##STR00026## ##STR00027## ##STR00028## ##STR00029##

##STR00030## ##STR00031## ##STR00032## ##STR00033## ##STR00034## ##STR00035##
##STR00036## ##STR00037## ##STR00038## ##STR00039## ##STR00040## ##STR00041##
##STR00042## ##STR00043## ##STR00044## ##STR00045## ##STR00046## ##STR00047##
##STR00048## ##STR00049## ##STR00050## ##STR00051## ##STR00052## ##STR00053##
##STR00054## ##STR00055## ##STR00056##

##STR00057## ##STR00058## ##STR00059## ##STR00060## ##STR00061## ##STR00062##
##STR00063## ##STR00064## ##STR00065## ##STR00066## ##STR00067## ##STR00068##
##STR00069## ##STR00070## ##STR00071## ##STR00072## ##STR00073## ##STR00074##
##STR00075## ##STR00076## ##STR00077## ##STR00078## ##STR00079## ##STR00080##
##STR00081## ##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086##
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##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103## ##STR00104##
##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109## ##STR00110##
##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115## ##STR00116##
##STR00117## ##STR00118## ##STR00119## ##STR00120## ##STR00121## ##STR00122##
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##STR00145## ##STR00146## ##STR00147## ##STR00148## ##STR00149## ##STR00150##
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##STR00191## ##STR00192## ##STR00193## ##STR00194## ##STR00195## ##STR00196##
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##STR00209## ##STR00210## ##STR00211## ##STR00212## ##STR00213##
##STR00214## ##STR00215## ##STR00216## ##STR00217## ##STR00218## ##STR00219##
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##STR00274## ##STR00275## ##STR00276## ##STR00277## ##STR00278## ##STR00279##
##STR00280## ##STR00281## ##STR00282##
##STR00283## ##STR00284## ##STR00285## ##STR00286## ##STR00287## ##STR00288##
##STR00289## ##STR00290## ##STR00291## ##STR00292## ##STR00293## ##STR00294##
##STR00295## ##STR00296## ##STR00297## ##STR00298## ##STR00299## ##STR00300##
##STR00301## ##STR00302## ##STR00303## ##STR00304## ##STR00305## ##STR00306##
##STR00307## ##STR00308## ##STR00309## ##STR00310## ##STR00311## ##STR00312##
##STR00313## ##STR00314## ##STR00315## ##STR00316## ##STR00317## ##STR00318##
##STR00319## ##STR00320## ##STR00321## ##STR00322## ##STR00323## ##STR00324##
##STR00325##
##STR00326## ##STR00327## ##STR00328## ##STR00329## ##STR00330## ##STR00331##
##STR00332## ##STR00333## ##STR00334## ##STR00335## ##STR00336## ##STR00337##
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##STR00363## ##STR00364## ##STR00365## ##STR00366## ##STR00367## ##STR00368##
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##STR00387## ##STR00388## ##STR00389##
##STR00390## ##STR00391## ##STR00392## ##STR00393## ##STR00394## ##STR00395##
##STR00396## ##STR00397##

[0123] In formula (1), Z^{sup.}+ is an onium cation, preferably sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2).

##STR00398##

[0124] In formulae (cation-1) and (cation-2), R^{sup.}ct1 to R^{sup.}ct5 are each independently halogen or a C_{sub}.1-C_{sub}.30 hydrocarbyl group which may contain a heteroatom. Suitable halogen atoms include fluorine, chlorine, bromine and iodine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C_{sub}.1-C_{sub}.30 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl; C_{sub}.3-C_{sub}.30 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, adamantyl; C_{sub}.2-C_{sub}.30 alkenyl groups such as vinyl, allyl, propenyl, butenyl, hexenyl; C_{sub}.3-C_{sub}.30 cyclic

unsaturated hydrocarbyl groups such as cyclohexenyl; C.sub.6-C.sub.30 aryl groups such as phenyl, naphthyl, thienyl; C.sub.7-C.sub.30 aralkyl groups such as benzyl, 1-phenylethyl, 2-phenylethyl, and combinations thereof. Inter alia, the aryl groups are preferred. In the hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, nitro, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0125] Also, R.sup.ct1 and R.sup.ct2 may bond together to form a ring with the sulfur atom to which they are attached. Exemplary structures of the ring are shown below.

##STR00399##

[0126] The broken line designates a point of attachment to R.sup.ct3.

[0127] Examples of the sulfonium cation having formula (cation-1) are shown below, but not limited thereto.

##STR00400## ##STR00401## ##STR00402## ##STR00403## ##STR00404## ##STR00405##
##STR00406## ##STR00407## ##STR00408## ##STR00409## ##STR00410## ##STR00411##
##STR00412## ##STR00413## ##STR00414## ##STR00415## ##STR00416## ##STR00417##
##STR00418## ##STR00419## ##STR00420## ##STR00421## ##STR00422##
##STR00423## ##STR00424## ##STR00425## ##STR00426## ##STR00427## ##STR00428##
##STR00429## ##STR00430## ##STR00431## ##STR00432## ##STR00433## ##STR00434##
##STR00435## ##STR00436## ##STR00437## ##STR00438## ##STR00439## ##STR00440##
##STR00441##
##STR00442## ##STR00443## ##STR00444## ##STR00445## ##STR00446## ##STR00447##
##STR00448##

[0128] Examples of the iodonium cation having formula (cation-2) are shown below, but not limited thereto.

##STR00449## ##STR00450## ##STR00451## ##STR00452##

[0129] Examples of the onium salt include arbitrary combinations of anions with cations, both as exemplified above.

[0130] The onium salt having formula (1) can be synthesized by well-known methods. For example, a method of preparing an onium salt having formula (PAG-1-ex) is described.

##STR00453##

[0131] Herein R.sup.1 to R.sup.13, R.sup.F, R.sup.15, L.sup.A, X.sup.L, m1 to m3, and Z.sup.+ are as defined above, M.sup.+ is a counter cation, and X.sup.— is a counter anion.

[0132] The first step is to react reactant SM-1 with reactant SM-2, which are commercially available or can be synthesized by any well-known synthesis method, to form an Intermediate In-1. A variety of condensing agents may be used when an ester bond is directly formed from the carboxy group on reactant SM-1 and the hydroxy group on reactant SM-2. Examples of the condensing agent used herein include N,N'-dicyclohexylcarbodiimide, N,N'-diisopropylcarbodiimide, 1-[3-(dimethylamino)propyl]-3-ethylcarbodiimide, and 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride. From the standpoint of ease of removal of urea compounds formed as by-products after reaction, it is desirable to use 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride. The reaction is carried out by dissolving reactants SM-1 and SM-2 in a halogenated solvent such as methylene chloride and adding a condensing agent. The reaction rate can be increased by adding 4-dimethylaminopyridine (DMAP) as a catalyst. While it is desirable in view of yield to monitor the reaction by silica gel thin layer chromatography (TLC) until the reaction is complete, the reaction time is typically about 12 to 24 hours. After the reaction is terminated, urea compounds formed by side reactions are removed by filtration or water-washing if necessary. The reaction solution is subjected to standard aqueous

work-up, obtaining Intermediate In-1. If necessary, the intermediate is purified by a standard technique such as chromatography or recrystallization.

[0133] The second step is a salt exchange between Intermediate In-1 and reactant SM-3 in the form of an onium salt: Z.sup.+X.sup.- to synthesize an onium salt PAG-1-ex. X.sup.- is preferably a chloride ion, bromide ion, iodide ion or methylsulfate anion because the exchange reaction readily takes place in a quantitative way. It is desirable in view of yield to monitor the reaction process by TLC. The reaction mixture is subjected to standard aqueous work-up, obtaining an onium salt PAG-1-ex. If necessary, the onium salt is purified by a standard technique such as chromatography or recrystallization.

[0134] In the above-illustrated scheme, the second step of ion exchange may be readily carried out by any well-known procedure, for example, with reference to JP-A 2007-145797.

[0135] It is noted that the preparation method according to the above scheme is merely exemplary and the method of preparing the inventive onium salt is not limited thereto.

[0136] The inventive onium salt is structurally characterized in that a substituted fused ring structure and an aromatic sulfonic acid structure are introduced in the anion. The substituted fused ring structure has a large excluded volume and thus serves as a bulky substituent to effectively inhibit the generated acid from diffusing. The substituted fused ring structure is also enough resistant to alkaline developer to prevent the pattern in unexposed region from losing a film thickness. The aromatic sulfonic acid structure, on the other hand, generates an acid of robust structure, which is effective for inhibiting acid diffusion. It is preferred that the aromatic ring that forms the aromatic sulfonic acid structure contain fluorine or an electron attractive sulfonate ester bond as a linking group. Then the generated acid has a higher acidity enough to efficiently deprotect the acid labile groups in the base polymer. Since fluorine has a high EUV absorption effect, which is not so high as iodine, the inclusion of more fluorine atoms leads to a greater number of secondary electrons, which promotes decomposition of cations and contributes to a sensitivity increase. JP 7109178 proposes an alkanesulfonic acid type photoacid generator having 2 to 4 fluorine atoms, which has the problems of noticeable acid diffusion because of alkanesulfonic acid and poor solvent solubility which leaves the risk of development defects. By virtue of the synergy of these effects, the resist composition comprising the inventive onium salt can form a resist pattern having a high sensitivity, low acid diffusion, and improved LWR of line patterns or improved CDU of hole patterns. The pattern is fully resistant to collapse. The resist composition is effective for forming small-size patterns.

[0137] The inventive onium salt is advantageously used as a PAG.

[Chemically Amplified Resist Composition]

(A) Photoacid Generator

[0138] Another embodiment of the invention is a chemically amplified resist composition essentially comprising (A) a photoacid generator in the form of the onium salt having formula (1).

[0139] In the chemically amplified resist composition, the amount of the onium salt having formula (1) as component (A) is preferably 0.1 to 40 parts by weight, more preferably 0.5 to 30 parts by weight per 80 parts by weight of a base polymer to be described just below. As long as the amount of component (A) is in the range, good sensitivity and resolution are achievable and the risk of foreign particles being formed after development or during stripping of resist film is avoided. The onium salt may be used alone or in admixture as component (A).

(B) Base Polymer

[0140] The chemically amplified resist composition may comprise a base polymer as component (B). The base polymer (B) contains repeat units having the formula (a1), which are also referred to as repeat units (a1).

##STR00454##

[0141] In formula (a1), R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl.

[0142] In formula (a1), X.sup.1 is a single bond, phenylene group, naphthylene group or *—C(=O)

—O—X^{sup.11}—. The phenylene or naphthylene group may be substituted with an optionally fluorinated C_{sub.1}-C_{sub.10} alkoxy moiety or halogen. X^{sup.11} is a C_{sub.1}-C_{sub.10} saturated hydrocarbylene group, phenylene or naphthylene group, and the saturated hydrocarbylene group may contain a hydroxy moiety, ether bond, ester bond or lactone ring. The asterisk (*) designates a point of attachment to the carbon atom in the backbone.

[0143] In formula (a1), AL^{sup.1} is an acid labile group. Examples of the acid labile group include those described in U.S. Pat. No. 8,574,817 (JP-A 2013-080033) and U.S. Pat. No. 8,846,303 (JP-A 2013-083821).

[0144] Typical of the acid labile group are groups of the following formulae (AL-1) to (AL-3).
##STR00455##

[0145] In formulae (AL-1) and (AL-2), R^{sup.L1} and R^{sup.L2} are each independently a C_{sub.1}-C_{sub.40} hydrocarbyl group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Inter alia, C_{sub.1}-C_{sub.20} hydrocarbyl groups are preferred.

[0146] In formula (AL-1), k is an integer of 0 to 10, preferably 1 to 5.

[0147] In formula (AL-2), R^{sup.L3} and R^{sup.L4} are each independently hydrogen or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Any two of R^{sup.L2}, R^{sup.L3} and R^{sup.L4} may bond together to form a C_{sub.3}-C_{sub.20} ring with the carbon atom or carbon and oxygen atoms to which they are attached. The ring preferably contains 4 to 16 carbon atoms and is typically alicyclic.

[0148] In formula (AL-3), R^{sup.L5}, R^{sup.L6} and R^{sup.L7} are each independently a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom such as oxygen, sulfur, nitrogen or fluorine. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Any two of R^{sup.L5}, R^{sup.L6} and R^{sup.L7} may bond together to form a C_{sub.3}-C_{sub.20} ring with the carbon atom to which they are attached. The ring preferably contains 4 to 16 carbon atoms and is typically alicyclic.

[0149] Examples of repeat unit (a1) are shown below, but not limited thereto. Herein R^{sup.A} and AL^{sup.1} are as defined above.

##STR00456## ##STR00457## ##STR00458## ##STR00459## ##STR00460## ##STR00461##

[0150] The base polymer (B) may further contain repeat units having the formula (a2), which are also referred to as repeat units (a2).

##STR00462##

[0151] In formula (a2), R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl. X^{sup.2} is a single bond or *—C(=O)—O— wherein * designates a point of attachment to the carbon atom in the backbone. R^{sup.21} is halogen, cyano, a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, a C_{sub.1}-C_{sub.20} hydrocarbyloxy group which may contain a heteroatom, a C_{sub.2}-C_{sub.20} hydrocarbylcarbonyl group which may contain a heteroatom, a C_{sub.2}-C_{sub.20} hydrocarbylcarbonyloxy group which may contain a heteroatom, or a C_{sub.2}-C_{sub.20} hydrocarbyloxycarbonyl group which may contain a heteroatom. AL^{sup.2} is an acid labile group, examples of which are as exemplified for the acid labile group AL. The subscript “a” is an integer of 0 to 4, preferably 0 or 1.

[0152] Examples of the repeat unit (a2) are shown below, but not limited thereto. Herein R^{sup.A} and AL^{sup.2} are as defined above.

##STR00463## ##STR00464## ##STR00465## ##STR00466## ##STR00467##

[0153] In a preferred embodiment, the base polymer further comprises repeat units having the formula (b1) or repeat units having the formula (b2), which are simply referred to as repeat units (b1) or (b2).

##STR00468##

[0154] In formulae (b1) and (b2), R^{sup.A} is each independently hydrogen, fluorine, methyl or

trifluoromethyl. Y.sup.1 is a single bond or *—C(=O)—O— wherein * designates a point of attachment to the carbon atom in the backbone. R.sup.31 is hydrogen or a C.sub.1-C.sub.20 group containing at least one structure selected from among hydroxy other than phenolic hydroxy, cyano, carbonyl, carboxy, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring and carboxylic anhydride (—C(=O)—O—C(=O)—). R.sup.32 is halogen, hydroxy, nitro, a C.sub.1-C.sub.20 hydrocarbonyl group which may contain a heteroatom, C.sub.1-C.sub.20 hydrocarbyloxy group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyl group which may contain a heteroatom, C.sub.2-C.sub.20 hydrocarbylcarbonyloxy group which may contain a heteroatom or C.sub.2-C.sub.20 hydrocarbyloxycarbonyl group which may contain a heteroatom. The subscript b is an integer of 1 to 4, c is an integer of 0 to 4, and b+c is from 1 to 5. [0155] Examples of the repeat unit (b1) are shown below, but not limited thereto. Herein, R.sup.A is as defined above.

##STR00469## ##STR00470## ##STR00471## ##STR00472## ##STR00473## ##STR00474##
##STR00475## ##STR00476## ##STR00477## ##STR00478## ##STR00479## ##STR00480##
##STR00481## ##STR00482## ##STR00483## ##STR00484## ##STR00485## ##STR00486##
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##STR00488## ##STR00489## ##STR00490## ##STR00491## ##STR00492## ##STR00493##
##STR00494## ##STR00495## ##STR00496## ##STR00497## ##STR00498## ##STR00499##
##STR00500## ##STR00501##

[0156] Examples of the repeat unit (b2) are shown below, but not limited thereto. Herein, R.sup.A is as defined above.

##STR00502## ##STR00503## ##STR00504## ##STR00505## ##STR00506## ##STR00507##
##STR00508## ##STR00509## ##STR00510##

[0157] Of the repeat units (b1) and (b2), those units having a lactone ring as the polar group are preferred in the ArF lithography and those units having a phenolic site are preferred in the KrF, EB and EUV lithography.

[0158] The base polymer may further comprise repeat units of at least one type selected from repeat units having the formulae (c1) to (c4), which are simply referred to as repeat units (c1) to (c4).

##STR00511##

[0159] In formulae (c1) to (c4), R is each independently hydrogen, fluorine, methyl or trifluoromethyl. Z.sup.1 is a single bond or phenylene group. Z.sup.2 is $\text{*—C(=O)—O—Z.sup.21—}$, $\text{*—C(=O)—N(H)—Z.sup.21—}$ or *—O—Z.sup.21— . Z.sup.21 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, a phenylene group or a divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety. Z.sup.3 is each independently a single bond, phenylene, naphthylene, or $\text{*—C(=O)—O—Z.sup.31—}$. Z.sup.31 is a C.sub.1-C.sub.10 aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, or phenylene or naphthylene group. Z.sup.4 is each independently a single bond, $\text{**—Z.sup.41—C(=O)—O—}$, $\text{**—C(=O)—NH—Z.sup.41—}$, or **—O—Z.sup.41— . Z.sup.41 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom. Z.sup.5 is each independently a single bond, $\text{*—Z.sup.51—C(=O)—O—}$, $\text{*—C(=O)—NH—Z.sup.51—}$, or *—O—Z.sup.51— . Z.sup.51 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom. Z.sup.6 is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, $\text{*—C(=O)—O—Z.sup.61—}$, $\text{*—C(=O)—N(H)—Z.sup.61—}$, or *—O—Z.sup.61— . Z.sup.61 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety. Herein, * designates a point of attachment to the carbon atom in the backbone, and ** designates a point of attachment to Z.sup.3.

[0160] The aliphatic hydrocarbylene group represented by Z.sup.21, Z.sup.31 and Z.sup.61 may be

straight, branched or cyclic. Examples thereof include alkanediyl groups such as methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,1-diyl, propane-1,2-diyl, propane-1,3-diyl, propane-2,2-diyl, butane-1,1-diyl, butane-1,2-diyl, butane-1,3-diyl, butane-2,3-diyl, butane-1,4-diyl, 1,1-dimethylethane-1,2-diyl, pentane-1,5-diyl, 2-methylbutane-1,2-diyl, and hexane-1,6-diyl; cycloalkanediyl groups such as cyclopropanediyl, cyclobutanediyl, cyclopentanediy and cyclohexanediyl, and combinations thereof.

[0161] The hydrocarbylene group represented by Z.sup.41 and Z.sup.51 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are shown below, but not limited thereto.

##STR00512##

[0162] In formula (c1), R.sup.41 and R.sup.42 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.20 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, and tert-butyl; C.sub.3-C.sub.20 cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclopentyl, cyclohexyl, cyclopropylmethyl, 4-methylcyclohexyl, cyclohexylmethyl, norbornyl, and adamantyl; C.sub.2-C.sub.20 alkenyl groups such as vinyl, allyl, propenyl, butenyl, and hexenyl; C.sub.3-C.sub.20 cyclic unsaturated hydrocarbyl groups such as cyclohexenyl; C.sub.6-C.sub.20 aryl groups such as phenyl, naphthyl and thienyl; C.sub.7-C.sub.20 aralkyl groups such as benzyl, 1-phenylethyl, and 2-phenylethyl, and combinations thereof. Of these, aryl groups are preferred. In the hydrocarbyl group, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0163] R.sup.41 and R.sup.42 may bond together to form a ring with the sulfur atom to which they are attached. Examples of the ring are as exemplified above for the ring that R.sup.ct1 and R.sup.ct2 in formula (cation-1) form with the sulfur atom to which they are attached.

[0164] Examples of the cation in repeat unit (c1) are given below, but not limited thereto. Herein R.sup.A is as defined above.

##STR00513## ##STR00514## ##STR00515## ##STR00516## ##STR00517## ##STR00518##
##STR00519## ##STR00520## ##STR00521## ##STR00522## ##STR00523## ##STR00524##
##STR00525## ##STR00526##

[0165] In formula (c1), M.sup.- is a non-nucleophilic counter ion. Halide, sulfonate, imide and methide anions are preferred. Examples of the non-nucleophilic counter ion include halide ions such as chloride and bromide ions; sulfonate anions, specifically fluoroalkylsulfonate ions such as triflate, 1,1,1-trifluoroethanesulfonate, and nonafluorobutanesulfonate, arylsulfonate ions such as tosylate, benzenesulfonate, 4-fluorobenzenesulfonate, and 1,2,3,4,5-pentafluorobenzenesulfonate, alkylsulfonate ions such as mesylate and butanesulfonate; imide ions such as bis(trifluoromethylsulfonyl)imide, bis(perfluoroethylsulfonyl)imide and bis(perfluorobutylsulfonyl)imide; and methide ions such as tris(trifluoromethylsulfonyl)methide and tris(perfluoroethylsulfonyl)methide.

[0166] Anions having the following formulae (ci-1) to (ci-4) are also useful as the non-nucleophilic counter ion.

##STR00527##

[0167] In formula (c1-1), R.sup.fa is fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as will be exemplified below for the hydrocarbyl group R.sup.fa1 in formula (c1-1-1).

[0168] Of the anions of formula (c1-1), an anion having the formula (c1-1-1) is preferred.

##STR00528##

[0169] In formula (c1-1-1), Q.sup.11 and Q.sup.12 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred for solvent solubility that at least one of Q.sup.11 and Q.sup.12 be trifluoromethyl. The subscript m is an integer of 0 to 4, most preferably 1.

[0170] R.sup.fa1 is a C.sub.1-C.sub.35 hydrocarbyl group which may contain a heteroatom. As the heteroatom, oxygen, nitrogen, sulfur and halogen atoms are preferred, with oxygen being most preferred. Of the hydrocarbyl groups, those groups of 6 to 30 carbon atoms are preferred from the aspect of achieving a high resolution in forming patterns of small feature size. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.35 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, pentyl, neopentyl, hexyl, heptyl, 2-ethylhexyl, nonyl, undecyl, tridecyl, pentadecyl, heptadecyl, and icosyl; C.sub.3-C.sub.35 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, 1-adamantyl, 2-adamantyl, 1-adamantylmethyl, norbornyl, norbornylmethyl, tricyclodecyl, tetracyclododecyl, tetracyclododecylmethyl, and dicyclohexylmethyl; C.sub.2-C.sub.35 unsaturated aliphatic hydrocarbyl groups such as allyl and 3-cyclohexenyl; C.sub.6-C.sub.35 aryl groups such as phenyl, 1-naphthyl, 2-naphthyl and 9-fluorenyl; and C.sub.7-C.sub.35 aralkyl groups such as benzyl and diphenylmethyl, and combinations thereof.

[0171] In the foregoing hydrocarbyl groups, some or all hydrogen may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, or some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, nitro, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Examples of the heteroatom-containing hydrocarbyl group include tetrahydrofuryl, methoxymethyl, ethoxymethyl, methylthiomethyl, acetamidomethyl, trifluoroethyl, (2-methoxyethoxy)methyl, acetoxymethyl, 2-carboxy-1-cyclohexyl, 2-oxopropyl, 4-oxo-1-adamantyl, and 3-oxocyclohexyl.

[0172] In formula (c1-1-1), L.sup.a1 is a single bond, ether bond, ester bond, sulfonate ester bond, carbonate bond or carbamate bond. From the aspect of synthesis, an ether bond or ester bond is preferred, with the ester bond being more preferred.

[0173] Examples of the anion having formula (c1-1) are shown below, but not limited thereto. Herein Q.sup.11 is as defined above.

##STR00529## ##STR00530## ##STR00531## ##STR00532## ##STR00533## ##STR00534##
##STR00535## ##STR00536## ##STR00537## ##STR00538## ##STR00539##

[0174] In formula (c1-2), R.sup.fb1 and R.sup.fb2 are each independently fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl group R.sup.fa1 in formula (c1-1-1). Preferably R.sup.fb1 and R.sup.fb2 are fluorine or C.sub.1-C.sub.4 straight fluorinated alkyl groups. Also, R.sup.fb1 and R.sup.fb2 may bond together to form a ring with the linkage: —CF.sub.2—SO.sub.2—N—SO.sub.2—CF.sub.2— to which they are attached. It is preferred that a combination of R.sup.fb1 and R.sup.fb2 be a fluorinated ethylene or fluorinated propylene group.

[0175] In formula (c1-3), R.sup.fc1, R.sup.fc2 and R.sup.fc3 are each independently fluorine or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbyl group R.sup.fc1 in formula (c1-1-1). Preferably R.sup.fc1, R.sup.fc2 and R.sup.fc3 are fluorine or C.sub.1-C.sub.4 straight fluorinated alkyl groups. Also, R.sup.fc1 and R.sup.fc2 may bond together to form a ring with the linkage: —CF.sub.2—SO.sub.2—C.sup.—

SO.sub.2—CF.sub.2— to which they are attached. It is preferred that a combination of R.sup.fc1 and R.sup.fc2 be a fluorinated ethylene or fluorinated propylene group.

[0176] In formula (c1-4), R.sup.fd is a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for R.sup.fa1.

[0177] Examples of the anion having formula (c1-4) are shown below, but not limited thereto.

##STR00540## ##STR00541##

[0178] Anions having an iodized or brominated aromatic ring are also useful as the non-nucleophilic counter ion. These anions have the formula (c1-5).

##STR00542##

[0179] In formula (c1-5), x is an integer of 1 to 3, y is an integer of 1 to 5, z is an integer of 0 to 3, and y+z is from 1 to 5. Preferably, y is 1, 2 or 3, more preferably 2 or 3, and z is 0, 1 or 2. [0180] X.sup.BI is iodine or bromine. A plurality of X.sup.BI may be identical or different when x and/or y is 2 or more. [0181] L.sup.11 is a single bond, ether bond, ester bond, or a C.sub.1-C.sub.6 saturated hydrocarbylene group which may contain an ether bond or ester bond. The saturated hydrocarbylene group may be straight, branched or cyclic. [0182] L.sup.12 is a single bond or a C.sub.1-C.sub.20 divalent linking group when x=1, or a C.sub.1-C.sub.20 (x+1)-valent linking group when x=2 or 3. The linking group may contain an oxygen, sulfur or nitrogen atom. [0183] R.sup.fe is hydroxy, carboxy, fluorine, chlorine, bromine, amino group, or a C.sub.1-C.sub.20 hydrocarbyl, C.sub.1-C.sub.20 hydrocarbyloxy, C.sub.2-C.sub.20 hydrocarbylcarbonyl, C.sub.2-C.sub.20 hydrocarbyloxycarbonyl, C.sub.2-C.sub.20 hydrocarbylcarbonyloxy, or C.sub.1-C.sub.20 hydrocarbylsulfonyloxy group, which may contain fluorine, chlorine, bromine, hydroxy, amino or ether bond, or —N(R.sup.feA)(R.sup.feB), —N(R.sup.feC)—C(=O)—R.sup.feD or —N(R.sup.feC)—C(=O)—O—R.sup.feA and R.sup.feB are each independently hydrogen or a C.sub.1-C.sub.6 saturated hydrocarbyl group. R.sup.feC is hydrogen, or a C.sub.1-C.sub.6 saturated hydrocarbyl group which may contain halogen, hydroxy, C.sub.1-C.sub.6 saturated hydrocarbyloxy, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyl or C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy moiety. R.sup.feD is a C.sub.1-C.sub.16 aliphatic hydrocarbyl group, C.sub.6-C.sub.12 aryl group or C.sub.7-C.sub.15 aralkyl group, which may contain halogen, hydroxy, C.sub.1-C.sub.6 saturated hydrocarbyloxy, C.sub.2-C.sub.6 saturated hydrocarbylcarbonyl or C.sub.2-C.sub.6 saturated hydrocarbylcarbonyloxy moiety. The aliphatic hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. The hydrocarbyl, hydrocarbyloxy, hydrocarbylcarbonyl, hydrocarbyloxycarbonyl, hydrocarbylcarbonyloxy, and hydrocarbylsulfonyloxy groups may be straight, branched or cyclic. A plurality of Re may be identical or different when x and/or z is 2 or more.

[0184] Of these, R.sup.fe is preferably hydroxy, —N(R.sup.feC)—C(=O)—R.sup.feD

N(R.sup.feC)—C(=O)—O—R.sup.feD, fluorine, chlorine, bromine, methyl or methoxy. [0185]

Rf.sup.11 to Rf.sup.14 are each independently hydrogen, fluorine or trifluoromethyl, at least one of Rf.sup.11 to Rf.sup.14 is fluorine or trifluoromethyl. Rf.sup.11 and Rf.sup.12, taken together, may form a carbonyl group. More preferably, both Rf.sup.13 and Rf.sup.14 are fluorine.

[0186] Examples of the anion having formula (c1-5) are shown below, but not limited thereto.

X.sup.BI is as defined above.

##STR00543## ##STR00544## ##STR00545## ##STR00546## ##STR00547## ##STR00548##

##STR00549## ##STR00550## ##STR00551## ##STR00552## ##STR00553## ##STR00554##

##STR00555## ##STR00556## ##STR00557## ##STR00558## ##STR00559## ##STR00560##

##STR00561## ##STR00562## ##STR00563## ##STR00564## ##STR00565## ##STR00566##

##STR00567## ##STR00568## ##STR00569## ##STR00570## ##STR00571## ##STR00572##

##STR00573## ##STR00574## ##STR00575## ##STR00576## ##STR00577##

##STR00578## ##STR00579## ##STR00580## ##STR00581## ##STR00582## ##STR00583##

##STR00584## ##STR00585## ##STR00586## ##STR00587## ##STR00588## ##STR00589##

##STR00590## ##STR00591## ##STR00592## ##STR00593## ##STR00594##

##STR00595## ##STR00596## ##STR00597##

[0187] Other useful examples of the non-nucleophilic counter ion include fluorobenzenesulfonic acid anions having an iodized aromatic ring bonded thereto as described in JP 6648726, anions having an acid-catalyzed decomposition mechanism as described in WO 2021/200056 and JP-A 2021-070692, anions having a cyclic ether group as described in JP-A 2018-180525 and JP-A 2021-035935, and anions as described in JP-A 2018-092159.

[0188] Further useful examples of the non-nucleophilic counter ion include bulky fluorine-free benzenesulfonic acid anions as described in JP-A 2006-276759, JP-A 2015-117200, JP-A 2016-065016, and JP-A 2019-202974; fluorine-free benzenesulfonic acid or alkylsulfonic acid anions having an iodized aromatic group bonded thereto as described in JP 6645464.

[0189] Also useful are bissulfonic acid anions as described in JP-A 2015-206932, sulfonamide or sulfonimide anions having sulfonic acid side and different side as described in WO 2020/158366, and anions having a sulfonic acid side and a carboxylic acid side as described in JP-A 2015-024989.

[0190] In formulae (c2) and (c3), L.sup.1 is a single bond, ether bond, ester bond, carbonyl, sulfonate ester bond, carbonate bond or carbamate bond. From the aspect of synthesis, an ether bond, ester bond or carbonyl is preferred, with the ester bond or carbonyl being more preferred.

[0191] In formula (c2), Rf.sup.1 and Rf.sup.2 are each independently fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred that both Rf.sup.1 and Rf.sup.2 be fluorine because the generated acid has a higher acid strength. Rf and Rf.sup.4 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is preferred for solvent solubility that at least one of Rf.sup.3 and Rf.sup.4 be trifluoromethyl.

[0192] In formula (c3), Rf.sup.5 and Rf.sup.6 are each independently hydrogen, fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group. It is excluded that all Rf.sup.5 and Rf.sup.6 are hydrogen at the same time. It is preferred for solvent solubility that at least one of Rf.sup.5 and Rf.sup.6 be trifluoromethyl.

[0193] In formulae (c2) and (c3), d is an integer of 0 to 3, preferably 1.

[0194] Examples of the anion in repeat unit (c2) are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00598## ##STR00599## ##STR00600## ##STR00601## ##STR00602## ##STR00603##
##STR00604## ##STR00605## ##STR00606## ##STR00607## ##STR00608## ##STR00609##
##STR00610## ##STR00611## ##STR00612## ##STR00613## ##STR00614## ##STR00615##
##STR00616## ##STR00617## ##STR00618## ##STR00619## ##STR00620## ##STR00621##
##STR00622##

[0195] Examples of the anion in repeat unit (63) are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00623## ##STR00624## ##STR00625## ##STR00626## ##STR00627## ##STR00628##
##STR00629## ##STR00630## ##STR00631## ##STR00632## ##STR00633## ##STR00634##
##STR00635## ##STR00636## ##STR00637## ##STR00638## ##STR00639## ##STR00640##
##STR00641## ##STR00642## ##STR00643## ##STR00644## ##STR00645## ##STR00646##

[0196] Examples of the anion in repeat unit (c4) are shown below, but not limited thereto. R.sup.A is as defined above.

##STR00647## ##STR00648##

[0197] In formulae (c2) to (c4), A.sup.+ is an onium cation. Suitable onium cations include ammonium, sulfonium and iodonium cations, with the sulfonium and iodonium cations being preferred. Examples thereof are as exemplified above for the sulfonium cation having formula (cation-1) and the iodonium cation having formula (cation-2) and as will be exemplified later for the ammonium cation having formula (cation-3).

[0198] Exemplary structures of the repeat units (c1) to (c4) include arbitrary combinations of

anions with cations, both as exemplified above.

[0199] Of the repeat units (c1) to (c4), repeat units (c2), (c3) and (c4) are preferred from the aspect of controlling acid diffusion. Repeat units (c2) and (c4) are more preferred from the aspect of the acid strength of generated acid. Repeat units (c2) are most preferred from the aspect of solvent solubility.

[0200] The base polymer may further comprise repeat units (d) of a structure having a hydroxy group protected with an acid labile group. The repeat unit (d) is not particularly limited as long as the unit includes one or more structures having a hydroxy group protected with a protective group such that the protective group is decomposed to generate the hydroxy group under the action of acid. Repeat units having the formula (d1) are preferred.

##STR00649##

[0201] In formula (d1), R^{sup}.A is as defined above. R^{sup}.41 is a C₁₋₃₀ (e+1)-valent hydrocarbon group which may contain a heteroatom. R^{sup}.42 is an acid labile group, and e is an integer of 1 to 4.

[0202] In formula (d1), the acid labile group R^{sup}.42 is deprotected under the action of acid so that a hydroxy group is generated. The structure of R^{sup}.42 is not particularly limited, an acetal structure, ketal structure, alkoxycarbonyl group and alkoxymethyl group having the following formula (d2) are preferred, with the alkoxymethyl group having formula (d2) being more preferred.

##STR00650##

[0203] Herein R^{sup}.43 is a C₁₋₁₅ hydrocarbonyl group.

[0204] Illustrative examples of the acid labile group R^{sup}.42, the alkoxymethyl group having formula (d2), and the repeat units (d) are as exemplified for the repeat units (d) in JP-A 2020-111564 (US 20200223796).

[0205] In addition to the foregoing units, the base polymer may further comprise repeat units (e) derived from indene, benzofuran, benzothiophene, acenaphthylene, chromone, coumarin, norbornadiene and derivatives thereof. Examples of the monomer from which repeat units (e) are derived are shown below, but not limited thereto.

##STR00651##

[0206] The base polymer may further comprise repeat units (f) derived from indane, vinylpyridine or vinylcarbazole.

[0207] In the polymer, repeat units (a1), (a2), (b1), (b2), (c1) to (c4), (d), (e), and (f) are incorporated in a ratio of preferably $0 < a1 \leq 0.8$, $0 \leq a2 \leq 0.8$, $0 \leq b1 \leq 0.6$, $0 \leq b2 \leq 0.6$, $0 \leq c1 \leq 0.4$, $0 \leq c2 \leq 0.4$, $0 \leq c3 \leq 0.4$, $0 \leq c4 \leq 0.4$, $0 \leq d \leq 0.5$, $0 \leq e \leq 0.3$, and $0 \leq f \leq 0.3$; more preferably $0 \leq a1 \leq 0.7$, $0 \leq a2 \leq 0.7$, $0 \leq b1 \leq 0.5$, $0 \leq b2 \leq 0.5$, $0 \leq c1 \leq 0.3$, $0 \leq c2 \leq 0.3$, $0 \leq c3 \leq 0.3$, $0 \leq c4 \leq 0.3$, $0 \leq d \leq 0.3$, $0 \leq e \leq 0.3$, and $0 \leq f \leq 0.3$.

[0208] The polymer should preferably have a weight average molecular weight (M_w) in the range of 1,000 to 500,000, and more preferably 3,000 to 100,000, as measured by GPC versus polystyrene standards using tetrahydrofuran (THF) or N,N-dimethylformamide (DMF) solvent. A M_w in the range ensures that the resist film has etch resistance and eliminates the risk of resolution decline by a failure to provide a difference in dissolution rate before and after exposure.

[0209] The polymer should preferably have a narrow dispersity (M_w/M_n) of 1.0 to 2.0 in order to provide a resist composition suitable for micropatterning to a small feature size for the reason that the influence of M_w/M_n becomes stronger as the pattern rule becomes finer. A M_w/M_n in the range ensures that the contents of lower and higher molecular weight polymer fractions are low and eliminates a possibility that foreign matter is left on the pattern or the pattern profile is degraded.

[0210] The polymer may be synthesized, for example, by dissolving a monomer or monomers corresponding to the above-mentioned repeat units in an organic solvent, adding a radical polymerization initiator, and heating for polymerization.

[0211] Examples of the organic solvent which can be used for polymerization include toluene, benzene, THF, diethyl ether, dioxane, cyclohexane, cyclopentane, methyl ethyl ketone (MEK),

propylene glycol monomethyl ether acetate (PGMEA), and γ -butyrolactone (GBL). Examples of the polymerization initiator used herein include 2,2'-azobisisobutyronitrile (AIBN), 2,2'-azobis(2,4-dimethylvaleronitrile), dimethyl 2,2'-azobis(2-methylpropionate), 1,1'-azobis(1-acetoxy-1-phenylethane), benzoyl peroxide, and lauroyl peroxide. The initiator is preferably added in an amount of 0.01 to 25 mol % based on the total of monomers to be polymerized. The reaction temperature is preferably 50 to 150° C., more preferably 60 to 100° C. The reaction time is preferably 2 to 24 hours, more preferably 2 to 12 hours in view of production efficiency.

[0212] The polymerization initiator may be fed to the reactor either by adding the initiator to the monomer solution and feeding the solution to the reactor, or by dissolving the initiator in a solvent to form an initiator solution and feeding the initiator solution and the monomer solution independently to the reactor. Because of a possibility that in the standby duration, the initiator generates a radical which triggers polymerization reaction to form a ultra-high-molecular-weight polymer, it is preferred from the standpoint of quality control to prepare the monomer solution and the initiator solution separately and add them dropwise. The acid labile group that has been incorporated in the monomer may be kept as such, or polymerization may be followed by protection or partial protection. During the polymer synthesis, any known chain transfer agent such as dodecyl mercaptan or 2-mercaptoethanol may be added for molecular weight control purpose. The amount of chain transfer agent added is preferably 0.01 to 20 mol % based on the total of monomers.

[0213] When a hydroxy-containing monomer is copolymerized, the hydroxy group is substituted by an acetal group which is susceptible to deprotection with acid, typically ethoxyethoxy, prior to polymerization, and the polymerization is followed by deprotection with weak acid and water. Alternatively, the hydroxy group is substituted by an acetyl, formyl or pivaloyl group prior to polymerization, and the polymerization is followed by alkaline hydrolysis.

[0214] When hydroxystyrene or hydroxyvinyl naphthalene is copolymerized, one method is dissolving hydroxystyrene or hydroxyvinyl naphthalene and other monomers in an organic solvent, adding a radical polymerization initiator thereto, and heating the solution for polymerization. In an alternative method, acetoxystyrene or acetoxyvinyl naphthalene is used instead, and after polymerization, the acetoxy group is deprotected by alkaline hydrolysis, for thereby converting the polymer product to polyhydroxystyrene or polyhydroxyvinyl naphthalene.

[0215] For alkaline hydrolysis, a base such as aqueous ammonia or triethylamine may be used. Preferably the reaction temperature is -20° C. to 100° C., more preferably 0° C. to 60° C., and the reaction time is 0.2 to 100 hours, more preferably 0.5 to 20 hours.

[0216] The amounts of monomers in the monomer solution may be determined appropriate so as to provide the preferred fractions of repeat units.

[0217] It is now described how to use the polymer obtained by the above preparation method. The reaction solution resulting from polymerization reaction may be used as the final product. Alternatively, the polymer may be recovered in powder form through a purifying step such as re-precipitation step of adding the polymerization solution to a poor solvent and letting the polymer precipitate as powder, after which the polymer powder is used as the final product. It is preferred from the standpoints of operation efficiency and consistent quality to handle a polymer solution which is obtained by dissolving the powder polymer resulting from the purifying step in a solvent, as the final product.

[0218] The solvents which can be used herein are described in JP-A 2008-111103, paragraphs [0144]-[0145] (U.S. Pat. No. 7,537,880). Exemplary solvents include ketones such as cyclohexanone and methyl-2-n-pentyl ketone; alcohols such as 3-methoxybutanol, 3-methyl-3-methoxybutanol, 1-methoxy-2-propanol, and 1-ethoxy-2-propanol; keto-alcohols such as diacetone alcohol (DAA); ethers such as propylene glycol monomethyl ether (PGME), ethylene glycol monomethyl ether, propylene glycol monoethyl ether, ethylene glycol monoethyl ether, propylene glycol dimethyl ether, and diethylene glycol dimethyl ether; esters such as propylene glycol

monomethyl ether acetate (PGMEA), propylene glycol monoethyl ether acetate, ethyl lactate, ethyl pyruvate, butyl acetate, methyl 3-methoxypropionate, ethyl 3-ethoxypropionate, tert-butyl acetate, tert-butyl propionate, and propylene glycol mono-tert-butyl ether acetate; lactones such as γ -butyrolactone (GBL); and high-boiling alcohols such as diethylene glycol, propylene glycol, glycerol, 1,4-butanediol, and 1,3-butanediol, which may be used alone or in admixture.

[0219] The polymer solution preferably has a polymer concentration of 0.01 to 30% by weight, more preferably 0.1 to 20% by weight.

[0220] Prior to use, the reaction solution or polymer solution is preferably filtered through a filter. Filtration is effective for consistent quality because foreign particles and gel which can cause defects are removed.

[0221] Suitable materials of which the filter is made include fluorocarbon, cellulose, nylon, polyester, and hydrocarbon base materials. Preferred for the filtration of a resist composition are filters made of fluorocarbons commonly known as Teflon®, hydrocarbons such as polyethylene and polypropylene, and nylon. While the pore size of the filter may be selected appropriate to comply with the desired cleanness, the filter preferably has a pore size of up to 100 nm, more preferably up to 20 nm. A single filter may be used or a plurality of filters may be used in combination. Although the filtering method may be single pass of the solution, preferably the filtering step is repeated by flowing the solution in a circulating manner. In the polymer preparation process, the filtering step may be carried out any times, in any order and in any stage. The reaction solution as polymerized or the polymer solution may be filtered, preferably both are filtered.

[0222] The base polymer (B) may be used alone or as a blend of two or more polymers which differ in compositional ratio, Mw and/or Mw/Mn. Component (B) may also be a blend of the base polymer defined above and a hydrogenated product of ROMP. For the ROMP, reference is made to JP-A 2003-066612.

(C) Organic Solvent

[0223] The resist composition may comprise an organic solvent as component (C). The organic solvent (C) is not particularly limited as long as the foregoing and other components are soluble therein. Suitable solvents include ketones such as cyclopentanone, cyclohexanone, and methyl-2-n-pentyl ketone; alcohols such as 3-methoxybutanol, 3-methyl-3-methoxybutanol, 1-methoxy-2-propanol, and 1-ethoxy-2-propanol; keto-alcohols such as diacetone alcohol (DAA); ethers such as propylene glycol monomethyl ether (PGME), ethylene glycol monomethyl ether, propylene glycol monoethyl ether, ethylene glycol monoethyl ether, propylene glycol dimethyl ether, and diethylene glycol dimethyl ether; esters such as propylene glycol monomethyl ether acetate (PGMEA), propylene glycol monoethyl ether acetate, ethyl lactate, ethyl pyruvate, butyl acetate, methyl 3-methoxypropionate, ethyl 3-ethoxypropionate, tert-butyl acetate, tert-butyl propionate, and propylene glycol mono-tert-butyl ether acetate; and lactones such as γ -butyrolactone (GBL), and mixtures thereof.

[0224] Of the foregoing organic solvents, it is recommended to use 1-ethoxy-2-propanol, PGMEA, cyclohexanone, GBL, DAA and mixtures thereof because the base polymer (B) is most soluble therein.

[0225] The organic solvent (C) is preferably added in an amount of 200 to 5,000 parts by weight, and more preferably 400 to 3,500 parts by weight per 80 parts by weight of the base polymer (B). The organic solvent may be used alone or in admixture.

(D) Quencher

[0226] The resist composition may further comprise a quencher as component (D). As used herein, the term “quencher” refers to a compound capable of trapping the acid generated by a PAG in the resist composition to prevent the acid from diffusing to the unexposed region and to assist in forming the desired pattern.

[0227] Onium salts having the formulae (2) and (3) are useful as the quencher (D).

##STR00652##

[0228] In formula (2), R.sup.q1 is hydrogen or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom, exclusive of the hydrocarbyl group in which the hydrogen bonded to the carbon atom at α -position of the sulfo group is substituted by fluorine or fluoroalkyl. In formula (3), R.sup.q2 is hydrogen or a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom.

[0229] The C.sub.1-C.sub.40 hydrocarbyl group R.sup.q1 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.40 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, sec-butyl, tert-butyl, n-pentyl, tert-pentyl, n-hexyl, n-octyl, 2-ethylhexyl, n-nonyl, and n-decyl; C.sub.3-C.sub.40 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, cyclopentylmethyl, cyclopentylethyl, cyclopentylbutyl, cyclohexylmethyl, cyclohexylethyl, cyclohexylbutyl, norbornyl, tricyclo[5.2.1.0.sup.2,6]decyl, and adamantyl; C.sub.6-C.sub.40 aryl groups such as phenyl, naphthyl and anthracenyl, and combinations thereof. In these hydrocarbyl groups, some or all hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, or some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, fluorine, chlorine, bromine, iodine, cyano, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0230] Examples of the hydrocarbyl group R.sup.q2 include fluorinated saturated hydrocarbyl groups such as trifluoromethyl and trifluoroethyl and fluorinated aryl groups such as pentafluorophenyl and 4-trifluoromethylphenyl as well as exemplary groups as mentioned above for R.sup.q1.

[0231] Examples of the anion in the onium salt having formula (2) are shown below, but not limited thereto.

##STR00653## ##STR00654## ##STR00655## ##STR00656## ##STR00657## ##STR00658##

[0232] Examples of the anion in the onium salt having formula (3) are shown below, but not limited thereto.

##STR00659## ##STR00660## ##STR00661## ##STR00662## ##STR00663##

[0233] In formulae (2) and (3), Mq.sup.+ is an onium cation, which is preferably selected from sulfonium cations having formula (cation-1), iodonium cations having formula (cation-2), as defined above, and ammonium cations having the following formula (cation-3).

##STR00664##

[0234] In formula (cation-3), R.sup.ct6 to R.sup.ct9 are each independently a C.sub.1-C.sub.40 hydrocarbyl group which may contain a heteroatom. A pair of R.sup.ct6 and R.sup.ct7 may bond together to form a ring with the nitrogen atom to which they are attached. Examples of the hydrocarbyl group are as exemplified above for R.sup.ct1 to R.sup.ct5 in formulae (cation-1) and (cation-2).

[0235] Examples of the ammonium cation having formula (cation-3) are shown below, but not limited thereto.

##STR00665##

[0236] Examples of the onium salt having formula (2) or (3) include arbitrary combinations of anions with cations, both as exemplified above. These onium salts may be readily prepared by ion exchange reaction using any well-known organic chemistry technique. For the ion exchange reaction, reference may be made to JP-A 2007-145797, for example.

[0237] The onium salt having formula (2) or (3) functions as a quencher in the chemically amplified resist composition because the counter anion of the onium salt is a conjugated base of a weak acid. As used herein, the weak acid indicates an acidity insufficient to deprotect an acid labile group from an acid labile group-containing unit in the base polymer. The onium salt having formula (2) or (3) functions as a quencher when used in combination with an onium salt type PAG having a conjugated base of a strong acid (typically α -fluorinated sulfonic acid) as the counter

anion. In a system using a mixture of an onium salt capable of generating a strong acid (typically α -fluorinated sulfonic acid) and an onium salt capable of generating a weak acid (typically non-fluorinated sulfonic acid or carboxylic acid), if the strong acid generated from the PAG upon exposure to high-energy radiation collides with the unreacted onium salt having a weak acid anion, then a salt exchange occurs whereby the weak acid is released and an onium salt having a strong acid anion is formed. In this course, the strong acid is exchanged into the weak acid having a low catalysis, incurring apparent deactivation of the acid for enabling to control acid diffusion.

[0238] Also useful as the quencher (D) are onium salts having sulfonium cation and phenoxide anion sites in a common molecule as described in JP 6848776, onium salts having sulfonium cation and carboxylate anion sites in a common molecule as described in JP 6583136 and JP-A 2020-200311, and onium salts having iodonium cation and carboxylate anion sites in a common molecule as described in JP 6274755.

[0239] If a PAG capable of generating a strong acid is an onium salt, an exchange from the strong acid generated upon exposure to high-energy radiation to a weak acid can take place as mentioned above, but it rarely happens that the weak acid generated upon exposure to high-energy radiation collides with the unreacted onium salt capable of generating a strong acid to induce a salt exchange. This is because of the phenomenon that an onium cation is more likely to form an ion pair with a stronger acid anion.

[0240] When the onium salt having formula (2) or (3) is used as the quencher (D), the amount of the onium salt used is preferably 0.1 to 20 parts by weight, more preferably 0.1 to 10 parts by weight per 80 parts by weight of the base polymer (B). As long as the amount of onium salt type quencher (D) is in the range, a satisfactory resolution is available without a substantial lowering of sensitivity. The onium salt having formula (2) or (3) may be used alone or in admixture.

[0241] Nitrogen-containing compounds may also be used as the quencher (D). Suitable nitrogen-containing compounds include primary, secondary and tertiary amine compounds, specifically amine compounds having a hydroxy group, ether bond, ester bond, lactone ring, cyano group or sulfonate ester bond, as described in JP-A 2008-111103, paragraphs [0146]-[0164] (U.S. Pat. No. 7,537,880), and primary or secondary amine compounds protected with a carbamate group, as described in JP 3790649.

[0242] A sulfonic acid sulfonium salt having a nitrogen-containing substituent may also be used as the nitrogen-containing compound. This compound functions as a quencher in the unexposed region, but as a so-called photo-degradable base in the exposed region because it loses the quencher function in the exposed region due to neutralization thereof with the acid generated by itself. Using a photo-degradable base, the contrast between exposed and unexposed regions can be further enhanced. With respect to the photo-degradable base, reference may be made to JP-A 2009-109595 and JP-A 2012-046501, for example.

[0243] When the nitrogen-containing compound is used as the quencher (D), the amount of the nitrogen-containing compound used is preferably 0.001 to 12 parts by weight, more preferably 0.01 to 8 parts by weight per 80 parts by weight of the base polymer (B). The nitrogen-containing compound may be used alone or in admixture.

(E) Other Photoacid Generator

[0244] The chemically amplified resist composition may comprise (E) a photoacid generator other than component (A). The other PAG is not particularly limited as long as it is capable of generating an acid upon exposure to high-energy radiation.

[0245] The preferred PAG is a salt having the formula (4) or (5).

##STR00666##

[0246] In formula (4), R.sup.101 to R.sup.105 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. Any two of R.sup.101, R.sup.102 and R.sup.103 may bond together to form a ring with the sulfur atom to which they are attached. Examples of the hydrocarbyl group are as exemplified above for R.sup.ct1 to R.sup.ct5 in formulae

(cation-1) and (cation-2).

[0247] Examples of the cation in the sulfonium salt having formula (4) are as exemplified above for the sulfonium cation having formula (cation-1). Examples of the cation in the iodonium salt having formula (5) are as exemplified above for the iodonium cation having formula (cation-2).

[0248] In formulae (4) and (5), Xa is an anion of strong acid selected from formulae (c1-1) to (ci-5).

[0249] Compounds having the formula (6) are also preferred as the other PAG (E).

##STR00667##

[0250] In formula (6), R.sup.201 and R.sup.202 are each independently a C.sub.1-C.sub.30 hydrocarbyl group which may contain a heteroatom. R.sup.203 is a C.sub.1-C.sub.30 hydrocarbylene group which may contain a heteroatom. Any two of R.sup.201, R.sup.202 and R.sup.203 may bond together to form a ring with the sulfur atom to which they are attached.

[0251] The C.sub.1-C.sub.30 hydrocarbyl group represented by R.sup.201 and R.sup.202 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.30 alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, tert-pentyl, n-hexyl, n-octyl, 2-ethylhexyl, n-nonyl, and n-decyl; C.sub.3-C.sub.30 cyclic saturated hydrocarbyl groups such as cyclopentyl, cyclohexyl, cyclopentylmethyl, cyclopentylethyl, cyclopentylbutyl, cyclohexylmethyl, cyclohexylethyl, cyclohexylbutyl, norbornyl, oxanorbornyl, tricyclo[5.2.1.0.sup.2,6]decyl, and adamantyl; and C.sub.6-C.sub.30 aryl groups such as phenyl, methylphenyl, ethylphenyl, n-propylphenyl, isopropylphenyl, n-butylphenyl, isobutylphenyl, sec-butylphenyl, tert-butylphenyl, naphthyl, methylnaphthyl, ethylnaphthyl, n-propylnaphthyl, isopropylnaphthyl, n-butylnaphthyl, isobutylnaphthyl, sec-butylnaphthyl, tert-butylnaphthyl, and anthracenyl, and combinations thereof. In these hydrocarbyl groups, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, and some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, cyano, fluorine, chlorine, bromine, iodine, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety.

[0252] The C.sub.1-C.sub.30 hydrocarbylene group R.sup.203 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include C.sub.1-C.sub.30 alkanediyl groups such as methanediyl, ethane-1,1-diyl, ethane-1,2-diyl, propane-1,3-diyl, butane-1,4-diyl, pentane-1,5-diyl, hexane-1,6-diyl, heptane-1,7-diyl, octane-1,8-diyl, nonane-1,9-diyl, decane-1,10-diyl, undecane-1,11-diyl, dodecane-1,12-diyl, tridecane-1,13-diyl, tetradecane-1,14-diyl, pentadecane-1,15-diyl, hexadecane-1,16-diyl, and heptadecane-1,17-diyl; C.sub.3-C.sub.30 cyclic saturated hydrocarbylene groups such as cyclopentanedyl, cyclohexanedyl, norbornanedyl and adamantanedyl; and arylene groups such as phenylene, methylphenylene, ethylphenylene, n-propylphenylene, isopropylphenylene, n-butylphenylene, isobutylphenylene, sec-butylphenylene, tert-butylphenylene, naphthylene, methylnaphthylene, ethylnaphthylene, n-propylnaphthylene, isopropylnaphthylene, n-butylnaphthylene, isobutylnaphthylene, sec-butylnaphthylene, and tert-butylnaphthylene. In these hydrocarbylene groups, some or all of the hydrogen atoms may be substituted by a moiety containing a heteroatom such as oxygen, sulfur, nitrogen or halogen, or some constituent —CH.sub.2— may be replaced by a moiety containing a heteroatom such as oxygen, sulfur or nitrogen, so that the group may contain a hydroxy, cyano, fluorine, chlorine, bromine, iodine, carbonyl, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, carboxylic anhydride (—C(=O)—O—C(=O)—) or haloalkyl moiety. Of the heteroatoms, oxygen is preferred.

[0253] In formula (6), L.sup.A is a single bond, ether bond or a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom. The hydrocarbylene group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for the hydrocarbylene

group R.sup.203.

[0254] In formula (6), X.sup.a, X.sup.b, X.sup.c and X.sup.d are each independently hydrogen, fluorine or trifluoromethyl, at least one of X.sup.a, X.sup.b, X.sup.c and X.sup.d being fluorine or trifluoromethyl.

[0255] Of the PAGs having formula (6), those having formula (6') are preferred.

##STR00668##

[0256] In formula (6'), L.sup.A is as defined above. X.sup.e is hydrogen or trifluoromethyl, preferably trifluoromethyl. R.sup.301, R.sup.302 and R.sup.303 are each independently hydrogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom. The hydrocarbyl group may be saturated or unsaturated and straight, branched or cyclic. Examples thereof are as exemplified above for Ra in formula (c1-1-1). The subscripts p and q are each independently an integer of 0 to 5, and r is an integer of 0 to 4.

[0257] Examples of the PAG having formula (6) include those exemplified for the PAG having formula (2) in JP-A 2017-026980.

[0258] Of the foregoing PAGs, those having an anion of formula (c1-1-1) or (c1-4) are especially preferred because of reduced acid diffusion and high solubility in solvents. Also those having formula (6') are especially preferred because of extremely reduced acid diffusion.

[0259] When used, the other PAG (E) is preferably added in an amount of 0.1 to 40 parts, and more preferably 0.5 to 20 parts by weight per 80 parts by weight of the base polymer (B). As long as the amount of the PAG is in the range, good resolution is achievable and the risk of foreign particles being formed after development or during stripping of resist film is avoided. The PAG may be used alone or in admixture.

(F) Surfactant

[0260] The resist composition may further include (F) a surfactant. Preferred are a surfactant which is insoluble or substantially insoluble in water and soluble in alkaline developer, and a surfactant which is insoluble or substantially insoluble in water and alkaline developer. For the surfactant, reference should be made to those compounds described in JP-A 2010-215608 and JP-A 2011-016746.

[0261] While many examples of the surfactant which is insoluble or substantially insoluble in water and alkaline developer are described in the patent documents cited herein, preferred examples are surfactants FC-4430 (3M), Olfine® E1004 (Nissin Chemical Co., Ltd.), Surflon® 5-381, KH-20 and KH-30 (AGC Seimi Chemical Co., Ltd.). Partially fluorinated oxetane ring-opened polymers having the formula (surf-1) are also useful.

##STR00669##

[0262] It is provided herein that R, Rf, A, B, C, m, and n are applied to only formula (surf-1), independent of their descriptions other than for the surfactant. R is a di- to tetra-valent C.sub.2-C.sub.5 aliphatic group. Exemplary divalent aliphatic groups include ethylene, 1,4-butylene, 1,2-propylene, 2,2-dimethyl-1,3-propylene and 1,5-pentylene. Exemplary tri- and tetra-valent groups are shown below.

##STR00670##

[0263] Herein the broken line denotes a valence bond. These formulae are partial structures derived from glycerol, trimethylol ethane, trimethylol propane, and pentaerythritol, respectively. Of these, 1,4-butylene and 2,2-dimethyl-1,3-propylene are preferably used.

[0264] Rf is trifluoromethyl or pentafluoroethyl, and preferably trifluoromethyl. The letter m is an integer of 0 to 3, n is an integer of 1 to 4, and the sum of m and n, which represents the valence of R, is an integer of 2 to 4. "A" is equal to 1, B is an integer of 2 to 25, and C is an integer of 0 to 10. Preferably, B is an integer of 4 to 20, and C is 0 or 1. Note that the formula (surf-1) does not prescribe the arrangement of respective constituent units while they may be arranged either blockwise or randomly. For the preparation of surfactants in the form of partially fluorinated oxetane ring-opened polymers, reference should be made to U.S. Pat. No. 5,650,483, for example.

[0265] The surfactant which is insoluble or substantially insoluble in water and soluble in alkaline developer is useful when ArF immersion lithography is applied to the resist composition in the absence of a resist protective film. In this embodiment, the surfactant has a propensity to segregate on the resist surface for achieving a function of minimizing water penetration or leaching. The surfactant is also effective for preventing water-soluble components from being leached out of the resist film for minimizing any damage to the exposure tool. The surfactant becomes solubilized during aqueous alkaline development following exposure and PEB, and thus forms few or no foreign particles which become defects. The preferred surfactant is a polymeric surfactant which is insoluble or substantially insoluble in water, but soluble in alkaline developer, also referred to as "hydrophobic resin" in this sense, and especially which is water repellent and enhances water sliding.

[0266] Suitable polymeric surfactants include those containing repeat units of at least one type selected from the formulae (7A) to (7E).

##STR00671##

[0267] Herein, R^{sup}.B is hydrogen, fluorine, methyl or trifluoromethyl. W^{sup}.1 is —CH₂—, —CH₂CH₂— or —O—, or two separate —H. R^{sup}.s1 is each independently hydrogen or a C₁-C₁₀ hydrocarbyl group. R^{sup}.s2 is a single bond or a C₁-C₅ straight or branched hydrocarbylene group. R^{sup}.s3 is each independently hydrogen, a C₁-C₁₅ hydrocarbyl or fluorinated hydrocarbyl group, or an acid labile group. When R^{sup}.s3 is a hydrocarbyl or fluorinated hydrocarbyl group, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond. R^{sup}.s4 is a C₁-C₂₀ (u+1)-valent hydrocarbon or fluorinated hydrocarbon group, and u is an integer of 1 to 3. R^{sup}.s5 is each independently hydrogen or a group: —C(=O)—O—R^{sup}.sa wherein R^{sup}.sa is a C₁-C₂₀ fluorinated hydrocarbyl group. R^{sup}.s6 is a C₁-C₁₅ hydrocarbyl or fluorinated hydrocarbyl group in which an ether bond or carbonyl moiety may intervene in a carbon-carbon bond.

[0268] The hydrocarbyl group represented by R^{sup}.s1 may be straight, branched or cyclic and is preferably saturated. Examples thereof include C₁-C₁₀ alkyl groups such as methyl, ethyl, n-propyl, isopropyl, n-butyl, isobutyl, sec-butyl, tert-butyl, n-pentyl, n-hexyl, n-heptyl, n-octyl, n-nonyl, n-decyl, and C₃-C₁₀ cyclic saturated hydrocarbyl groups such as cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, adamantyl, and norbornyl. Inter alia, C₁-C₆ hydrocarbyl groups are preferred.

[0269] The hydrocarbylene group represented by R^{sup}.s2 may be straight, branched or cyclic and is preferably saturated. Examples thereof include methylene, ethylene, propylene, butylene and pentylene.

[0270] The hydrocarbyl group represented by R^{sup}.s3 or R^{sup}.s6 may be saturated or unsaturated and straight, branched or cyclic. Examples thereof include saturated hydrocarbyl groups, and aliphatic unsaturated hydrocarbyl groups such as alkenyl and alkynyl groups, with the saturated hydrocarbyl groups being preferred. Suitable saturated hydrocarbyl groups include those exemplified for the hydrocarbyl group represented by R^{sup}.s1 as well as undecyl, dodecyl, tridecyl, tetradecyl, and pentadecyl. Examples of the fluorinated hydrocarbyl group represented by R^{sup}.s3 or R^{sup}.s6 include the foregoing hydrocarbyl groups in which some or all carbon-bonded hydrogen atoms are substituted by fluorine atoms. In these groups, an ether bond or carbonyl moiety may intervene in a carbon-carbon bond as mentioned above.

[0271] Examples of the acid labile group represented by R^{sup}.s3 include groups of the above formulae (AL-3) to (AL-5), trialkylsilyl groups in which each alkyl moiety has 1 to 6 carbon atoms, and C₄-C₂₀ oxoalkyl groups.

[0272] The (u+1)-valent hydrocarbon or fluorinated hydrocarbon group represented by R^{sup}.s4 may be straight, branched or cyclic and examples thereof include the foregoing hydrocarbyl or fluorinated hydrocarbyl groups from which u number of hydrogen atoms are eliminated.

[0273] The fluorinated hydrocarbyl group represented by R^{sup}.sa may be straight, branched or

cyclic and is preferably saturated. Examples thereof include the foregoing hydrocarbyl groups in which some or all hydrogen atoms are substituted by fluorine atoms. Illustrative examples include trifluoromethyl, 2,2,2-trifluoroethyl, 3,3,3-trifluoro-1-propyl, 3,3,3-trifluoro-2-propyl, 2,2,3,3-tetrafluoropropyl, 1,1,1,3,3,3-hexafluoroisopropyl, 2,2,3,3,4,4,4-heptafluorobutyl, 2,2,3,3,4,4,5,5-octafluoropentyl, 2,2,3,3,4,4,5,5,6,6,7,7-dodecafluoroheptyl, 2-(perfluorobutyl)ethyl, 2-(perfluorohexyl)ethyl, 2-(perfluorooctyl)ethyl, and 2-(perfluorodecyl)ethyl.

[0274] Examples of the repeat units having formulae (7A) to (7E) are shown below, but not limited thereto. Herein R_{sup}.B is as defined above.

##STR00672## ##STR00673## ##STR00674## ##STR00675##

[0275] The polymeric surfactant may further contain repeat units other than the repeat units having formulae (7A) to (7E). Typical other repeat units are those derived from methacrylic acid and α -trifluoromethylacrylic acid derivatives. In the polymeric surfactant, the content of the repeat units having formulae (7A) to (7E) is preferably at least 20 mol %, more preferably at least 60 mol %, most preferably 100 mol % of the overall repeat units.

[0276] The polymeric surfactant preferably has a Mw of 1,000 to 500,000, more preferably 3,000 to 100,000 and a Mw/Mn of 1.0 to 2.0, more preferably 1.0 to 1.6.

[0277] The polymeric surfactant may be synthesized by any desired method, for example, by dissolving an unsaturated bond-containing monomer or monomers providing repeat units having formula (7A) to (7E) and optionally other repeat units in an organic solvent, adding a radical initiator, and heating for polymerization. Suitable organic solvents used herein include toluene, benzene, THF, diethyl ether, and dioxane. Examples of the polymerization initiator used herein include AIBN, 2,2'-azobis(2,4-dimethylvaleronitrile), dimethyl 2,2-azobis(2-methylpropionate), benzoyl peroxide, and lauroyl peroxide. Preferably the reaction temperature is 50 to 100° C. and the reaction time is 4 to 24 hours. The acid labile group that has been incorporated in the monomer may be kept as such, or the polymer may be protected or partially protected therewith at the end of polymerization.

[0278] During the synthesis of polymeric surfactant, any known chain transfer agent such as dodecyl mercaptan or 2-mercaptoethanol may be added for molecular weight control purpose. The amount of chain transfer agent added is preferably 0.01 to 10 mol % based on the total moles of monomers to be polymerized.

[0279] When the resist composition contains a surfactant (F), the amount thereof is preferably 0.1 to 50 parts by weight, and more preferably 0.5 to 10 parts by weight per 80 parts by weight of the base polymer (B). At least 0.1 part of the surfactant is effective in improving the receding contact angle with water of the resist film at its surface. Up to 50 parts of the surfactant is effective in forming a resist film having a low rate of dissolution in a developer and capable of maintaining the height of a small-size pattern formed therein. The surfactant (F) may be used alone or in admixture.

(G) Other Components

[0280] The resist composition may further comprise (G) another component, for example, a compound which is decomposed with an acid to generate another acid (i.e., acid amplifier compound), an organic acid derivative, a fluorinated alcohol, and a compound having a Mw of up to 3,000 which changes its solubility in developer under the action of an acid (i.e., dissolution inhibitor). Specifically, the acid amplifier compound is described in JP-A 2009-269953 and JP-A 2010-215608 and preferably used in an amount of 0 to 5 parts, more preferably 0 to 3 parts by weight per 80 parts by weight of the base polymer (B). An extra amount of the acid amplifier compound can make the acid diffusion control difficult and cause degradations to resolution and pattern profile. With respect to the remaining additives, reference should be made to JP-A 2009-269953 and JP-A 2010-215608.

[Process]

[0281] A further embodiment of the invention is a process of forming a pattern from the resist composition defined above by lithography. The preferred process includes the steps of applying the

resist composition onto a substrate to form a resist film thereon, exposing the resist film to high-energy radiation, and developing the exposed resist film in a developer. Any desired steps may be added to the process if necessary.

[0282] The substrate used herein may be a substrate for integrated circuitry fabrication, e.g., Si, SiO₂, SiN, SiON, TiN, WSi, BPSG, SOG, organic antireflective film, etc. or a substrate for mask circuitry fabrication, e.g., Cr, CrO, CrON, MoSi₂, SiO₂, etc.

[0283] The resist composition is applied onto a substrate by a suitable coating technique such as spin coating. The coating is prebaked on a hot plate preferably at a temperature of 60 to 150° C. for 1 to 10 minutes, more preferably at 80 to 140° C. for 1 to 5 minutes. The resulting resist film preferably has a thickness of 0.05 to 2 μm.

[0284] Then the resist film is exposed to a pattern of high-energy radiation, typically KrF or ArF excimer laser, EUV or EB. On use of KrF excimer laser, ArF excimer laser or EUV, the resist film is exposed through a mask having a desired pattern, preferably in a dose of 1 to 200 mJ/cm², more preferably 10 to 100 mJ/cm². On use of EB, a pattern may be written directly or through a mask having the desired pattern, preferably in a dose of 1 to 300 μC/cm², more preferably 10 to 200 μC/cm².

[0285] The exposure may be performed by conventional lithography whereas the immersion lithography of holding a liquid having a refractive index of at least 1.0 between the resist film and the projection lens may be employed if desired. The liquid is typically water, and in this case, a protective film which is insoluble in water may be formed on the resist film.

[0286] While the water-insoluble protective film serves to prevent any components from being leached out of the resist film and to improve water sliding on the film surface, it is generally divided into two types. The first type is an organic solvent-strippable protective film which must be stripped, prior to alkaline development, with an organic solvent in which the resist film is not dissolvable. The second type is an alkali-soluble protective film which is soluble in an alkaline developer so that it can be removed simultaneously with the removal of solubilized regions of the resist film. The protective film of the second type is preferably of a material comprising a polymer having a 1,1,1,3,3,3-hexafluoro-2-propanol residue (which is insoluble in water and soluble in an alkaline developer) as a base in an alcohol solvent of at least 4 carbon atoms, an ether solvent of 8 to 12 carbon atoms or a mixture thereof. Alternatively, the aforementioned surfactant which is insoluble in water and soluble in an alkaline developer may be dissolved in an alcohol solvent of at least 4 carbon atoms, an ether solvent of 8 to 12 carbon atoms or a mixture thereof to form a material from which the protective film of the second type is formed.

[0287] After the exposure, the resist film may be baked (PEB), for example, on a hotplate preferably at 60 to 150° C. for 1 to 5 minutes, more preferably at 80 to 140° C. for 1 to 3 minutes.

[0288] The resist film is then developed with a developer in the form of an aqueous base solution, for example, 0.1 to 5 wt %, preferably 2 to 3 wt % aqueous solution of tetramethylammonium hydroxide (TMAH) for 0.1 to 3 minutes, preferably 0.5 to 2 minutes by conventional techniques such as dip, puddle and spray techniques. In this way, the exposed region of the resist film is dissolved away, and a desired resist pattern is formed on the substrate.

[0289] Any desired step may be added to the pattern forming process. For example, after the resist film is formed, a step of rinsing with pure water may be introduced to extract the acid generator or the like from the film surface or wash away particles. After exposure, a step of rinsing may be introduced to remove any water remaining on the film after exposure.

[0290] Also, a double patterning process may be used for pattern formation. The double patterning process includes a trench process of processing an underlay to a 1:3 trench pattern by a first step of exposure and etching, shifting the position, and forming a 1:3 trench pattern by a second step of exposure, for forming a 1:1 pattern; and a line process of processing a first underlay to a 1:3 isolated left pattern by a first step of exposure and etching, shifting the position, processing a second underlay formed below the first underlay by a second step of exposure through the 1:3

isolated left pattern, for forming a half-pitch 1:1 pattern.

[0291] In the pattern forming process, negative tone development may also be used. That is, an organic solvent may be used instead of the aqueous alkaline solution as the developer for developing and dissolving away the unexposed region of the resist film.

[0292] The organic solvent used as the developer is preferably selected from 2-octanone, 2-nonanone, 2-heptanone, 3-heptanone, 4-heptanone, 2-hexanone, 3-hexanone, diisobutyl ketone, methylcyclohexanone, acetophenone, methylacetophenone, propyl acetate, butyl acetate, isobutyl acetate, pentyl acetate, isopentyl acetate, butenyl acetate, propyl formate, butyl formate, isobutyl formate, pentyl formate, isopentyl formate, methyl valerate, methyl pentenoate, methyl crotonate, ethyl crotonate, methyl propionate, ethyl propionate, ethyl 3-ethoxypropionate, methyl lactate, ethyl lactate, propyl lactate, butyl lactate, isobutyl lactate, pentyl lactate, isopentyl lactate, methyl 2-hydroxyisobutyrate, ethyl 2-hydroxyisobutyrate, methyl benzoate, ethyl benzoate, phenyl acetate, benzyl acetate, methyl phenylacetate, benzyl formate, phenylethyl formate, methyl 3-phenylpropionate, benzyl propionate, ethyl phenylacetate, and 2-phenylethyl acetate. These organic solvents may be used alone or in admixture of two or more.

EXAMPLES

[0293] Synthesis Examples, Examples and Comparative Examples are given below by way of illustration and not by way of limitation. The abbreviation “pbw” is parts by weight. Analysis is made by IR, ¹H-NMR spectroscopy and time-of-flight mass spectrometry using analytic instruments as shown below.

[0294] IR: NICOLET 6700 by Thermo Fisher Scientific Inc.

[0295] ¹H-NMR: ECA-500 by JEOL Ltd.

[0296] MALDI TOF-MS: S3000 by JEOL Ltd.

[1] Synthesis of Onium Salts

[Example 1-1] Synthesis of Onium Salt PAG-1

##STR00676##

(1) Synthesis of Intermediate In-1

[0297] In nitrogen atmosphere, a reactor was charged with 6.3 g of reactant SM-1, 10.4 g of reactant SM-2, 0.3 g of DMAP, and 50 g of methylene chloride and cooled in an ice bath. While the internal temperature of the reactor was kept below 20° C., 5.8 g of 1-ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride in powder form was added. Thereafter, the reactor was allowed to reach room temperature, at which the reaction solution was aged for 12 hours. At the end of aging, water was added to quench the reaction, followed by ordinary aqueous work-up. The subsequent steps of distilling off the solvent and washing the residue with diisopropyl ether gave 14.4 g of Intermediate In-1 as oily matter (yield 92%).

(2) Synthesis of Onium Salt PAG-1

[0298] In nitrogen atmosphere, a reactor was charged with 14.4 g of Intermediate In-1, 7.8 g of reactant SM-3, 50 g of methylene chloride, and 30 g of water, which were stirred for 15 minutes. The organic layer was taken out, washed with water, and concentrated under reduced pressure. To the concentrate, 50 g of methyl isobutyl ketone was added, followed by azeotropic dewatering. Diisopropyl ether was then added for recrystallization, obtaining 16.0 g of the target onium salt, PAG-1 as white crystals (yield 92%).

[0299] PAG-1 was analyzed by IR spectroscopy and TOF-MS, with the data shown below. FIG. 1 is the ¹H-NMR/DMSO-d₆ spectrum of PAG-1.

[0300] IR (D-ATR): ν =3484, 3065, 3024, 2972, 1779, 1641, 1593, 1583, 1479, 1467, 1443, 1379, 1324, 1272, 1229, 1170, 1123, 1091, 1047, 999, 986, 917, 886, 847, 816, 763, 706, 681, 631, 596, 535, 487 cm⁻¹.

[0301] MALDI TOF-MS: [0302] positive M⁺ 277 (corresponding to

C₁₈H₁₃OS⁺) [0303] negative M⁻ 477 (corresponding to

C₂₃H₁₃F₄O₅S⁻)

[Examples 1-2 to 1-9] Synthesis of Onium Salts PAG-2 to PAG-9
[0304] Onium salts PAG-2 to PAG-9 shown below were synthesized using corresponding reactants and well-known organic synthesis reactions.

##STR00677## ##STR00678##

[2] Synthesis of Base Polymers

[Synthesis Example] Synthesis of Base Polymers P-1 to P-5

[0305] Base polymers P-1 to P-5 were synthesized by combining monomers, performing copolymerization reaction in MEK solvent, pouring the reaction solution to hexane for precipitation, washing the solid precipitate with hexane, isolation and drying. The polymer was analyzed for composition by ¹H-NMR spectroscopy and for Mw and Mw/Mn by GPC versus polystyrene standards using THF solvent.

##STR00679## ##STR00680## ##STR00681##

[3] Preparation of Chemically Amplified Resist Compositions

Examples 2-1 to 2-30 and Comparative Examples 1-1 to 1-20

[0306] Chemically amplified resist compositions (R-1 to R-30, CR-1 to CR-20) in solution form were prepared by dissolving a photoacid generator in the form of onium salt (PAG-1 to PAG-9) or comparative photoacid generator (PAG-A to PAG-E), other photoacid generator (PAG-X, PAG-Y), base polymer (P-1 to P-5), and quencher (Q-1 to Q-4) in a solvent containing 0.01 wt % of surfactant A in accordance with the formulation shown in Tables 1 and 2, and filtering through a Teflon® filter with a pore size of 0.2 μm.

TABLE-US-00001 TABLE 1 Other Base Photoacid photoacid Resist polymer generator generator
Quencher Solvent 1 Solvent 2 composition (pbw) (pbw) (pbw) (pbw) (pbw) (pbw) Example 2-1 R-1 P-1 (80) PAG-1 (28) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-2 R-2 P-1 (80) PAG-2 (28) — Q-1 (7.8) PGMEA (2200) DAA (900) 2-3 R-3 P-1 (80) PAG-3 (29) — Q-1 (7.4) PGMEA (2200) DAA (900) 2-4 R-4 P-1 (80) PAG-4 (27) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-5 R-5 P-1 (80) PAG-5 (15) PAG-X (10) Q-1 (8.0) PGMEA (2200) DAA (900) 2-6 R-6 P-1 (80) PAG-6 (28) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-7 R-7 P-1 (80) PAG-7 (28) — Q-1 (7.8) PGMEA (2200) DAA (900) 2-8 R-8 P-1 (80) PAG-8 (29) — Q-1 (7.4) PGMEA (2200) DAA (900) 2-9 R-9 P-1 (80) PAG-9 (15) PAG-Y (8) Q-1 (8.0) PGMEA (2200) DAA (900) 2-10 R-10 P-2 (80) PAG-1 (29) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-11 R-11 P-2 (80) PAG-2 (30) — Q-3 (7.8) PGMEA (2200) DAA (900) 2-12 R-12 P-2 (80) PAG-4 (28) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-13 R-13 P-2 (80) PAG-5 (18) PAG-Y (7) Q-1 (8.0) PGMEA (2200) DAA (900) 2-14 R-14 P-2 (80) PAG-7 (28) — Q-2 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 2-15 R-15 P-3 (80) PAG-1 (10) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-16 R-16 P-3 (80) PAG-2 (9) — Q-3 (7.6) PGMEA (2200) DAA (900) 2-17 R-17 P-3 (80) PAG-5 (8) PAG-Y (4) Q-1 (7.8) PGMEA (2200) DAA (900) 2-18 R-18 P-3 (80) PAG-7 (9) — Q-3 (8.0) PGMEA (2200) DAA (900) 2-19 R-19 P-3 (80) PAG-9 (8) PAG-X (4) Q-2 (8.0) PGMEA (2200) DAA (900) 2-20 R-20 P-4 (80) PAG-1 (10) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-21 R-21 P-4 (80) PAG-3 (10) — Q-1 (7.6) PGMEA (2200) DAA (900) 2-22 R-22 P-4 (80) PAG-4 (8) PAG-X (4) Q-3 (8.0) PGMEA (2200) DAA (900) 2-23 R-23 P-4 (80) PAG-6 (10) — Q-2 (8.0) PGMEA (2200) DAA (900) 2-24 R-24 P-4 (80) PAG-8 (8) — Q-3 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 2-25 R-25 P-5 (80) PAG-1 (10) — Q-1 (8.0) PGMEA (2200) DAA (900) 2-26 R-26 P-5 (80) PAG-2 (8) — Q-2 (8.0) PGMEA (2200) DAA (900) 2-27 R-27 P-5 (80) PAG-4 (8) — Q-3 (7.8) PGMEA (2200) DAA (900) 2-28 R-28 P-5 (80) PAG-6 (10) — Q-1 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 2-29 R-29 P-5 (80) PAG-7 (10) — Q-3 (8.0) PGMEA (2200) DAA (900) 2-30 R-30 P-5 (80) PAG-9 (8) PAG-Y (4) Q-2 (7.6) PGMEA (2200) DAA (900)

TABLE-US-00002 TABLE 2 Other Base Photoacid photoacid Resist polymer generator generator
Quencher Solvent 1 Solvent 2 composition (pbw) (pbw) (pbw) (pbw) (pbw) (pbw) Comparative 1-1 CR-1 P-1 (80) PAG-A (28) — Q-1 (8.0) PGMEA (2200) DAA (900) Example 1-2 CR-2 P-1 (80) PAG-B (28) — Q-1 (7.8) PGMEA (2200) DAA (900) 1-3 CR-3 P-1 (80) PAG-C (29) — Q-1 (7.8) PGMEA (2200) DAA (900) 1-4 CR-4 P-1 (80) PAG-D (27) — Q-1 (8.2) PGMEA (2200) DAA

(900) 1-5 CR-5 P-1 (80) PAG-E (15) PAG-X (10) Q-1 (8.2) PGMEA (2200) DAA (900) 1-6 CR-6 P-2 (80) PAG-A (27) — Q-2 (8.0) PGMEA (2200) DAA (900) 1-7 CR-7 P-2 (80) PAG-B (25) — Q-1 (8.0) PGMEA (2200) DAA (900) 1-8 CR-8 P-2 (80) PAG-E (18) PAG-Y (7) Q-1 (8.0) PGMEA (2200) DAA (900) 1-9 CR-9 P-3 (80) PAG-A (10) — Q-1 (7.8) PGMEA (2200) DAA (900) 1-10 CR-10 P-3 (80) PAG-B (9) — Q-3 (8.0) PGMEA (2200) DAA (900) 1-11 CR-11 P-3 (80) PAG-C (10) — Q-1 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 1-12 CR-12 P-3 (80) PAG-D (8) — Q-2 (7.6) PGMEA (2200) DAA (900) 1-13 CR-13 P-4 (80) PAG-A (10) — Q-1 (7.8) PGMEA (2200) DAA (900) 1-14 CR-14 P-4 (80) PAG-B (10) — Q-2 (8.0) PGMEA (2200) DAA (900) 1-15 CR-15 P-4 (80) PAG-C (10) — Q-3 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 1-16 CR-16 P-4 (80) PAG-E (8) PAG-X (4) Q-1 (8.0) PGMEA (2200) DAA (900) 1-17 CR-17 P-5 (80) PAG-A (10) — Q-1 (8.0) PGMEA (2200) DAA (900) 1-18 CR-18 P-5 (80) PAG-B (10) — Q-3 (7.6) PGMEA (2200) DAA (900) 1-19 CR-19 P-5 (80) PAG-C (8) — Q-1 (4.0) PGMEA (2200) DAA (900) Q-4 (4.0) 1-20 CR-20 P-5 (80) PAG-E (8) PAG-Y (4) Q-2 (8.0) PGMEA (2200) DAA (900)

[0307] The solvents, other photoacid generators PAG-X and PAG-Y, comparative photoacid generators PAG-A to PAG-E, quenchers Q-1 to Q-4, and surfactant A in Tables 1 and 2 are identified below.

Solvent:

[0308] PGMEA (propylene glycol monomethyl ether acetate) [0309] DAA (diacetone alcohol)

[0310] Other photoacid generators: PAG-X and PAG-Y

##STR00682##

[0311] Comparative photoacid generators: PAG-A to PAG-E

##STR00683##

[0312] Quenchers: Q-1 to Q-4

##STR00684##

[0313] Surfactant A: 3-methyl-3-(2,2,2-trifluoroethoxymethyl)oxetane/tetrahydrofuran/2,2-dimethyl-1,3-propane diol copolymer (Omnova Solutions, Inc.)

##STR00685## [0314] $a:(b+b'):(c+c')=1:4-7:0.01-1$ (molar ratio) [0315] Mw=1,500

[4] EUV Lithography Test 1

Examples 3-1 to 3-30 and Comparative Examples 2-1 to 2-20

[0316] Each of the chemically amplified resist compositions (R-1 to R-30, CR-1 to CR-20) was spin coated on a silicon substrate having a 20-nm coating of silicon-containing spin-on hard mask SHB-A940 (Shin-Etsu Chemical Co., Ltd., silicon content 43 wt %) and prebaked on a hotplate at 100° C. for 60 seconds to form a resist film of 50 nm thick. Using an EUV scanner NXE3400 (ASML, NA 0.33, σ 0.9/0.6, dipole illumination), the resist film was exposed to EUV through a mask bearing a LS pattern having a size of 18 nm and a pitch of 36 nm (on-wafer size) while varying the dose and focus (dose pitch: 1 mJ/cm.sup.2, focus pitch: 0.020 μ m). The resist film was baked (PEB) on a hotplate at the temperature shown in Tables 3 and 4 for 60 seconds and puddle developed in a 2.38 wt % TMAH aqueous solution for 30 seconds, rinsed with a rinse fluid containing surfactant, and spin dried to form a positive pattern.

[0317] The LS pattern as developed was observed under CD-SEM (CG6300, Hitachi High-Technologies Corp.) whereupon sensitivity, EL, LWR, DOF and collapse limit were evaluated by the following methods. The results are shown in Tables 3 and 4.

[Evaluation of Sensitivity]

[0318] The optimum dose Eop (mJ/cm.sup.2) which provided a LS pattern with a line width of 18 nm and a pitch of 36 nm was determined as an index of sensitivity. A smaller value indicates a higher sensitivity.

[Evaluation of EL]

[0319] The exposure dose which provided a LS pattern with a space width of 18 nm $\pm$ 10% (i.e., 16.2 to 19.8 nm) was determined. EL (%) is calculated from the exposure doses according to the

following equation:

[00001]EL(%) = (.Math. $E_1 - E_2$.Math. / Eop) × 100 [0320] wherein E.sub.1 is an optimum exposure dose which provides a LS pattern with a line width of 16.2 nm and a pitch of 36 nm, E.sub.2 is an optimum exposure dose which provides a LS pattern with a line width of 19.8 nm and a pitch of 36 nm, and Eop is an optimum exposure dose which provides a LS pattern with a line width of 18 nm and a pitch of 36 nm. A greater value indicates better performance.

[Evaluation of LWR]

[0321] For the LS pattern formed by exposure at the optimum dose Eop, the line width was measured at 10 longitudinally spaced apart points, from which a 3-fold value (36) of the standard deviation (6) was determined and reported as LWR. A smaller value of 36 indicates a pattern having small roughness and uniform line width.

[Evaluation of DOF]

[0322] As an index of DOF, a range of focus which provided a LS pattern with a size of 18 nm±10% (i.e., 16.2 to 19.8 nm) was determined. A greater value indicates a wider DOF.

[Evaluation of Collapse Limit of Line Pattern]

[0323] For the LS pattern formed by exposure at the dose corresponding to the optimum focus, the line width was measured at 10 longitudinally spaced apart points. The minimum line size above which lines could be resolved without collapse was determined and reported as collapse limit. A smaller value indicates better collapse limit.

TABLE-US-00003 TABLE 3 PEB Collapse Resist temp. Eop EL LWR DOF limit composition (° C.) (mJ/cm.sup.2) (%) (nm) (nm) (nm) Example 3-1 R-1 100 36 19 2.6 120 10.4 3-2 R-2 100 37 17 2.5 120 10.7 3-3 R-3 95 37 18 2.6 120 10.6 3-4 R-4 105 37 17 2.6 110 10.8 3-5 R-5 105 36 18 2.7 100 10.9 3-6 R-6 95 37 19 2.8 110 11.1 3-7 R-7 105 38 19 2.6 120 11.2 3-8 R-8 105 37 18 2.7 120 11.3 3-9 R-9 100 37 19 2.7 120 10.9 3-10 R-10 95 36 17 2.7 120 10.8 3-11 R-11 100 38 18 2.7 100 10.8 3-12 R-12 105 37 19 2.8 110 11.1 3-13 R-13 105 37 18 2.9 110 10.8 3-14 R-14 100 36 17 2.7 110 10.8 3-15 R-15 100 37 18 2.6 120 10.6 3-16 R-16 100 36 19 2.7 120 10.7 3-17 R-17 105 38 18 2.6 110 11.2 3-18 R-18 95 37 19 2.7 120 10.8 3-19 R-19 100 37 17 2.7 110 10.8 3-20 R-20 105 36 18 2.7 100 11.3 3-21 R-21 95 36 19 2.8 120 11.2 3-22 R-22 100 37 17 2.7 110 10.7 3-23 R-23 105 36 18 2.6 120 11.4 3-24 R-24 95 36 19 2.8 120 11.1 3-25 R-25 100 37 18 2.7 110 10.9 3-26 R-26 95 35 17 2.8 120 10.7 3-27 R-27 100 37 18 2.7 100 10.8 3-28 R-28 100 36 19 2.8 120 10.8 3-29 R-29 100 36 17 2.7 110 11.1 3-30 R-30 95 37 19 2.9 110 11.2

TABLE-US-00004 TABLE 4 PEB Collapse Resist temp. Eop EL LWR DOF limit composition (° C.) (mJ/cm.sup.2) (%) (nm) (nm) (nm) Comparative 2-1 CR-1 100 38 14 3.0 80 11.8 Example 2-2 CR-2 100 39 14 3.2 90 12.3 2-3 CR-3 105 40 15 3.1 100 12.1 2-4 CR-4 100 40 15 3.5 80 12.1 2-5 CR-5 100 39 14 3.3 70 11.9 2-6 CR-6 100 39 13 3.2 80 12.3 2-7 CR-7 100 40 15 3.1 90 12.2 2-8 CR-8 105 39 14 3.1 80 12.6 2-9 CR-9 100 40 13 3.2 100 12.4 2-10 CR-10 100 41 13 3.3 90 12.7 2-11 CR-11 95 40 14 3.3 80 12.1 2-12 CR-12 100 40 15 3.1 80 12.3 2-13 CR-13 100 39 16 3.1 70 12.1 2-14 CR-14 100 39 15 3.0 90 11.9 2-15 CR-15 100 41 14 3.1 80 12.1 2-16 CR-16 95 40 15 3.2 90 11.9 2-17 CR-17 100 38 13 3.3 80 12.5 2-18 CR-18 105 40 15 3.2 80 11.8 2-19 CR-19 100 39 14 3.4 90 11.8 2-20 CR-20 100 38 15 3.2 80 12.1

[0324] It is demonstrated in Tables 3 and 4 that chemically amplified resist compositions comprising PAGs in the form of onium salts within the scope of the invention exhibit a high sensitivity and improved values of EL, LWR and DOF. Small values of collapse limit attest that in forming a small-size pattern, the pattern is resistant to collapse. The resist compositions are useful in the EUV lithography process.

[5] EUV Lithography Test 2

Examples 4-1 to 4-30 and Comparative Examples 3-1 to 3-20

[0325] Each of the chemically amplified resist compositions (R-1 to R-30, CR-1 to CR-20) was spin coated on a silicon substrate having a 20-nm coating of silicon-containing spin-on hard mask SHB-A940 (Shin-Etsu Chemical Co., Ltd., silicon content 43 wt %) and prebaked on a hotplate at

105° C. for 60 seconds to form a resist film of 50 nm thick. Using an EUV scanner NXE3400 (ASML, NA 0.33, σ 0.9/0.6, quadrupole illumination), the resist film was exposed to EUV through a mask bearing a hole pattern having a pitch of 46 nm $\pm$ 20% bias (on-wafer size). The resist film was baked (PEB) on a hotplate at the temperature shown in Tables 5 and 6 for 60 seconds and developed in a 2.38 wt % TMAH aqueous solution for 30 seconds to form a hole pattern having a size of 23 nm.

[0326] The pattern as developed was observed under CD-SEM (CG6300, Hitachi High-Technologies Corp.). The dose (mJ/cm.sup.2) at which a pattern with a hole size of 23 nm was printed was determined as an index of sensitivity. The size of 50 holes was measured, from which a 3-fold value (3 σ) of the standard deviation (σ) was determined as a dimensional variation (or CDU). The results are shown in Tables 5 and 6.

TABLE-US-00005 TABLE 5 PEB Optimum Resist temp. dose CDU composition (° C.)

(mJ/cm.sup.2)	(nm)	Example 4-1	R-1	95	23	2.3	4-2	R-2	95	23	2.4	4-3	R-3	95	24	2.5	4-4	R-4	90	24	2.3	4-5	R-5	90	24	2.5	4-6	R-6	95	25	2.6	4-7	R-7	90	24	2.6	4-8	R-8	95	23	2.4	4-9	R-9	90	24	2.5	4-10	R-10	95	25	2.4	4-11	R-11	90	25	2.4	4-12	R-12	90	25	2.5	4-13	R-13	95	24	2.4	4-14	R-14	95	25	2.5	4-15	R-15	95	23	2.4	4-16	R-16	95	25	2.5	4-17	R-17	90	25	2.4	4-18	R-18	90	24	2.5	4-19	R-19	95	24	2.3	4-20	R-20	90	26	2.4	4-21	R-21	95	24	2.5	4-22	R-22	90	25	2.4	4-23	R-23	95	24	2.4	4-24	R-24	90	23	2.6	4-25	R-25	95	25	2.4	4-26	R-26	90	25	2.5	4-27	R-27	95	24	2.5	4-28	R-28	90	24	2.4	4-29	R-29	90	25	2.4	4-30	R-30	95	24	2.5
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TABLE-US-00006 TABLE 6 PEB Optimum Resist temp. dose CDU composition (° C.)

(mJ/cm.sup.2)	(nm)	Comparative 3-1	CR-1	90	26	2.9	Example 3-2	CR-2	90	26	2.7	3-3	CR-3	85	27	3.1	3-4	CR-4	90	26	2.9	3-5	CR-5	95	25	3.1	3-6	CR-6	90	26	3.0	3-7	CR-7	85	27	2.9	3-8	CR-8	90	26	3.2	3-9	CR-9	90	27	2.8	3-10	CR-10	90	27	2.9	3-11	CR-11	85	28	3.1	3-12	CR-12	90	27	3.1	3-13	CR-13	90	27	3.0	3-14	CR-14	90	28	3.2	3-15	CR-15	90	29	2.8	3-16	CR-16	85	27	2.9	3-17	CR-17	95	26	2.9	3-18	CR-18	90	27	2.9	3-19	CR-19	90	28	2.8	3-20	CR-20	90	29	2.9
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[0327] It is demonstrated in Tables 5 and 6 that chemically amplified resist compositions comprising PAGs in the form of onium salts within the scope of the invention exhibit a high sensitivity and satisfactory CDU.

[0328] Japanese Patent Application No. 2024-031892 is incorporated herein by reference. Although some preferred embodiments have been described, many modifications and variations may be made thereto in light of the above teachings. It is therefore to be understood that the invention may be practiced otherwise than as specifically described without departing from the scope of the appended claims.

Claims

1. An onium salt having the formula (1): ##STR00686## wherein R.sup.1 to R.sup.12 are each independently hydrogen, halogen, or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, one of R.sup.13 and R.sup.14 is a group having a partial structure having the formula (1a) and the other is hydrogen, halogen or a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, at least two of R.sup.1 to R.sup.14 may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and Z.sup.+ is an onium cation, ##STR00687## wherein m1 is 0 or 1, m2 is an integer of 0 to 4 when m1=0 and an integer of 0 to 6 when m1=1, m3 is an integer of 0 to 3 when m1=0 and an integer of 0 to 5 when m1=1, m2+m3 is an integer of 0 to 4 when m1=0 and an integer of 0 to 6 when m1=1, R.sup.F is fluorine, a C.sub.1-C.sub.6 fluorinated alkyl group, C.sub.1-C.sub.6 fluorinated alkoxy group, or C.sub.1-C.sub.6 fluorinated alkylthio group, R.sup.15 is a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, a plurality of R.sup.15 may bond together to form a ring with the carbon atoms to which they are attached when m3 is 2 or more, L.sup.A and L.sup.B are each independently a single bond, ether bond, ester bond, amide bond,

sulfonate ester bond, sulfonamide bond, carbonate bond or carbamate bond, and X^{sup.1} is a single bond or a C_{sub.1}-C_{sub.40} hydrocarbylene group which may contain a heteroatom.

2. The onium salt of claim 1, having the formula (1A): ##STR00688## wherein R^{sup.1} to R^{sup.13}, R^{sup.15}, L^{sup.A}, X^{sup.L}, R^{sup.F}, m₁ to m₃, and Z^{sup.+} are as defined above.

3. The onium salt of claim 2, having the formula (1B): ##STR00689## wherein R^{sup.5}, R^{sup.10} to R^{sup.13}, R^{sup.15}, L^{sup.A}, X^{sup.L}, R^{sup.F}, m₁ to m₃, and Z^{sup.+} are as defined above, m₄ and m₅ are each independently an integer of 0 to 4, R^{sup.16} and R^{sup.17} are each independently hydrogen, halogen, or a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom; when m₄ is 2 or more, a plurality of R^{sup.16} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon, and when m₅ is 2 or more, a plurality of R^{sup.17} may bond together to form a ring with the carbon to which they are attached or the carbon to which they are attached and the intervenient carbon.

4. The onium salt of claim 3, having the formula (1C): ##STR00690## wherein R^{sup.5}, R^{sup.10} to R^{sup.13}, R^{sup.16}, R^{sup.17}, L^{sup.A}, X^{sup.L}, m₄, m₅, and Z^{sup.+} are as defined above.

5. The onium salt of claim 1 wherein Z^{sup.+} is a sulfonium cation having the formula (cation-1) or iodonium cation having the formula (cation-2): ##STR00691## wherein R^{sup.ct1} to R^{sup.ct5} are each independently halogen or a C_{sub.1}-C_{sub.30} hydrocarbyl group which may contain a heteroatom, R^{sup.ct1} and R^{sup.ct2} may bond together to form a ring with the sulfur atom to which they are attached.

6. A photoacid generator in the form of the onium salt of claim 1.

7. A chemically amplified resist composition comprising the photoacid generator of claim 6.

8. The resist composition of claim 7, comprising a base polymer comprising repeat units having the formula (a1): ##STR00692## wherein R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl, X^{sup.1} is a single bond, phenylene, naphthylene or *—C(=O)—O—X^{sup.11}—, the phenylene or naphthylene group may be substituted with an optionally fluorinated C_{sub.1}-C_{sub.10} alkoxy moiety or halogen, X^{sup.11} is a C_{sub.1}-C_{sub.10} saturated hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, a phenylene group or a naphthylene group, * designates a point of attachment to the carbon atom in the backbone, and AL^{sup.1} is an acid labile group.

9. The resist composition of claim 8 wherein the base polymer further comprises repeat units having the formula (a2): ##STR00693## wherein R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl, X^{sup.2} is a single bond or *—C(=O)—O—, * designates a point of attachment to the carbon atom in the backbone, R^{sup.21} is halogen, cyano, a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, C_{sub.1}-C_{sub.20} hydrocarbyloxy group which may contain a heteroatom, C_{sub.2}-C_{sub.20} hydrocarbylcarbonyl group which may contain a heteroatom, C_{sub.2}-C_{sub.20} hydrocarbylcarbonyloxy group which may contain a heteroatom, or C_{sub.2}-C_{sub.20} hydrocarbyloxycarbonyl group which may contain a heteroatom, AL^{sup.2} is an acid labile group, and a is an integer of 0 to 4.

10. The resist composition of claim 8 wherein the base polymer further comprises repeat units having the formula (b1) or (b2): ##STR00694## wherein R^{sup.A} is hydrogen, fluorine, methyl or trifluoromethyl, Y^{sup.1} is a single bond or *—C(=O)—O—, * designates a point of attachment to the carbon atom in the backbone, R^{sup.31} is hydrogen, or a C_{sub.1}-C_{sub.20} group containing at least one structure selected from hydroxy exclusive of phenolic hydroxy, cyano, carbonyl, carboxy, ether bond, ester bond, sulfonate ester bond, carbonate bond, lactone ring, sultone ring, and carboxylic anhydride (—C(=O)—O—C(=O)—), R^{sup.32} is halogen, hydroxy, nitro, a C_{sub.1}-C_{sub.20} hydrocarbyl group which may contain a heteroatom, C_{sub.1}-C_{sub.20} hydrocarbyloxy group which may contain a heteroatom, C_{sub.2}-C_{sub.20} hydrocarbylcarbonyl group which may contain a heteroatom, C_{sub.2}-C_{sub.20} hydrocarbylcarbonyloxy group which may contain a heteroatom, or C_{sub.2}-C_{sub.20} hydrocarbyloxycarbonyl group which may contain a heteroatom, b is an integer of 1 to 4, c is an integer of 0 to 4, and b+c is from 1 to 5.

11. The resist composition of claim 8 wherein the base polymer further comprises repeat units of at least one type selected from repeat units having the formula (c1), repeat units having the formula (c2), repeat units having the formula (c3), and repeat units having the formula (c4): ##STR00695## wherein R.sup.A is hydrogen, fluorine, methyl or trifluoromethyl, Z.sup.1 is a single bond or phenylene group, Z.sup.2 is *—C(=O)—O—Z.sup.21—, *—C(=O)—NH—Z.sup.21—, or *—O—Z.sup.21—, Z.sup.21 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group or a divalent group obtained by combining the foregoing, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, Z.sup.3 is each independently a single bond, phenylene, naphthylene, or *—C(=O)—O—Z.sup.31—, Z.sup.31 is a C.sub.1-C.sub.10 aliphatic hydrocarbylene group which may contain a hydroxy moiety, ether bond, ester bond or lactone ring, or phenylene or naphthylene group, Z.sup.4 is each independently a single bond, **—Z.sup.41—C(=O)—O—, **—C(=O)—NH—Z.sup.41—, or —O—Z.sup.41—, Z.sup.41 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, Z.sup.5 is each independently a single bond, *—Z.sup.5—C(=O)—O—, *—C(=O)—NH—Z.sup.5—, or *—O—Z.sup.5, Z.sup.51 is a C.sub.1-C.sub.20 hydrocarbylene group which may contain a heteroatom, Z.sup.6 is a single bond, methylene, ethylene, phenylene, fluorinated phenylene, trifluoromethyl-substituted phenylene, *—C(=O)—O—Z.sup.61—, *—C(=O)—N(H)—Z.sup.61—, or *—Z.sup.61—, Z.sup.61 is a C.sub.1-C.sub.6 aliphatic hydrocarbylene group, phenylene group, fluorinated phenylene group or trifluoromethyl-substituted phenylene group, which may contain a carbonyl moiety, ester bond, ether bond or hydroxy moiety, * designates a point of attachment to the carbon atom in the backbone, ** designates a point of attachment to Z.sup.3, R.sup.41 and R.sup.42 are each independently a C.sub.1-C.sub.20 hydrocarbyl group which may contain a heteroatom, R.sup.41 and R.sup.42 may bond together to form a ring with the sulfur atom to which they are attached, L.sup.1 is a single bond, ether bond, ester bond, carbonyl group, sulfonate ester bond, carbonate bond or carbamate bond, Rf.sup.1 and Rf.sup.2 are each independently fluorine or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, Rf.sup.3 and Rf.sup.4 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, Rf.sup.5 and Rf.sup.6 are each independently hydrogen, fluorine, or a C.sub.1-C.sub.6 fluorinated saturated hydrocarbyl group, excluding that all Rf.sup.5 and Rf.sup.6 are hydrogen at the same time, M.sup.— is a non-nucleophilic counter ion, A.sup.+ is an onium cation, and d is an integer of 0 to 3.

12. The resist composition of claim 7, further comprising an organic solvent.

13. The resist composition of claim 7, further comprising a quencher.

14. The resist composition of claim 7, further comprising a photoacid generator other than the photoacid generator.

15. The resist composition of claim 7, further comprising a surfactant.

16. A pattern forming process comprising the steps of applying the chemically amplified resist composition of claim 7 onto a substrate to form a resist film thereon, exposing the resist film to high-energy radiation, and developing the exposed resist film in a developer.

17. The pattern forming process of claim 16 wherein the high-energy radiation is KrF excimer laser radiation, ArF excimer laser radiation, EB or EUV of wavelength 3 to 15 nm.
